

MAGNETOM

Function Description System

Gradient

Document Version

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1.1 General

The **K2259 Gradient System** consists of switch-mode current generators (amplifiers) and an actively shielded gradient coil. The power capabilities of both these components have reached new levels unparalleled in industry until now.

1.2 Gradient Amplifier Specifications

The K2259 Gradient System is largely equivalent to the K2217 gradients of the existing MAGNETOM Symphony and Sonata systems. All components were redesigned to achieve even higher performance.

The performance values for the available gradient option is listed in the table below:

Tab. 1 Gradient options and Performance values

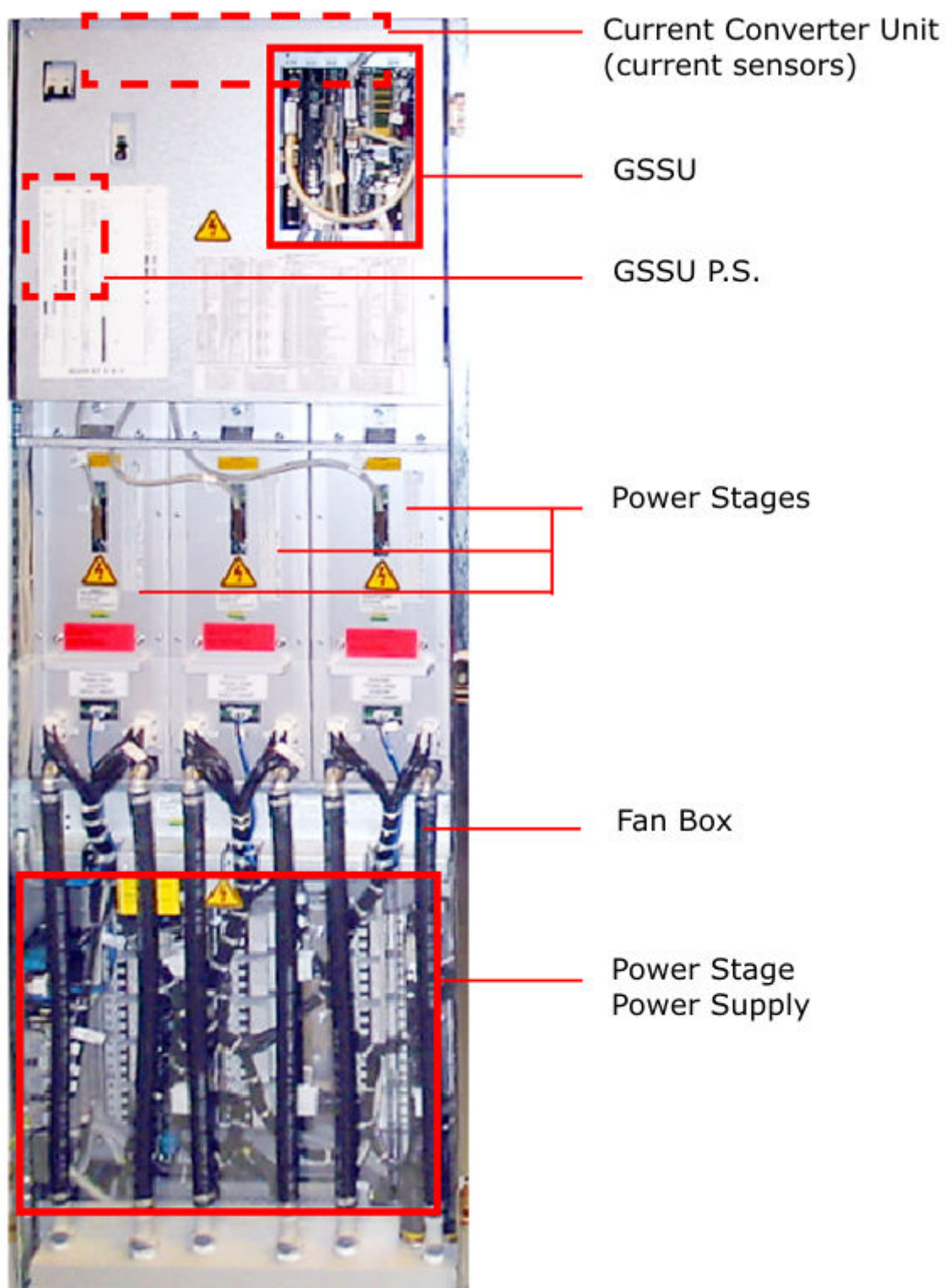
Performance	TQ
System type	Trio
Max. gradient field (X/Y/Z)	40/40/45 mT/m
Min. rise time	200 µsec. (0 - 40 mT/m)
Slew rate	200 T/m/s
Max. gradient current	625 A
Max. voltage (across Gradient Coil)	2000 V
Gradient Coil	AS 092

1.3 Gradient Amplifier Components

The Gradient System hardware components consist of eight functional blocks:

- Gradient Small Signal Unit (GSSU)
- GSSU Power Supply (GSSU-PS)
- Power Stage Supply
- Power Stage (3x)
- Output Filter Chokes
- Fan Box
- Current Converter Unit (CCU)
- Gradient Coil Assembly

Fig. 1: Gradient System K2259



1.3.1 Gradient Small Signal Unit (GSSU)

The task of the GSSU is to control the function of the Gradient Power Amplifier (GPA), generating signals required to drive a precise current in the Gradient Coils.

Additionally, the GSSU provides a CAN-bus communication link and signal feedback to the MR system for service loop tests and Tune-up procedures.

1.3.2 GSSU Power Supply

The GSSU Power Supply provides stabilized DC voltages (+5 V and ± 15 V) for the GSSU, 17 VDC for the driver transistor supply found inside the Power Stages and 24 VDC for the relays in the Power Stage Supply section and the Fan Box.

1.3.3 Power Stage (cascade design)

The Power Stage (PS) is a switch-mode current generator based on power IGBT arrays. In order to allow for a compact construction, the power devices are mounted on water-cooled heat sinks. Each Power Stage box consists of five so-called "H-bridges" that are connected in series (cascaded) to increase the output voltage. One "H-bridge" consists of four switches (IGBT modules) required to generate a bipolar output current.

1.3.4 Output Filter Chokes

The switching ripple on the Power Stages output current is eliminated by six filter chokes located behind the Power Stages. Per axis, two of them are connected in series to the Gradient Coil. To access the chokes, the Power Stages have to be removed.

1.3.5 Fan Box

For cooling the Power Stages and output filter chokes, five 24 VDC fans are located in a box underneath. Each fan is equipped with a built-in monitoring circuitry.

1.3.6 Current Converter Unit (CCU)

The Current Converter Unit implements current sensors manufactured by DANFYSIK and LEM.

The DANFYSIK sensors are responsible for providing accurate feedback signals proportional to the actual current driven through the Gradient Coils, which are required for regulation while maintaining galvanic decoupling from the output current.

The LEMs, based on the principle of Hall detection, are only used for monitoring functions and are of lesser performance.

1.3.7 Power Stage Supply

The Power Stage Supply consists of a dedicated transformer and rectifier circuits for the five cascades of each Power Stage. Each of the five cascades is supplied with 400 VDC.

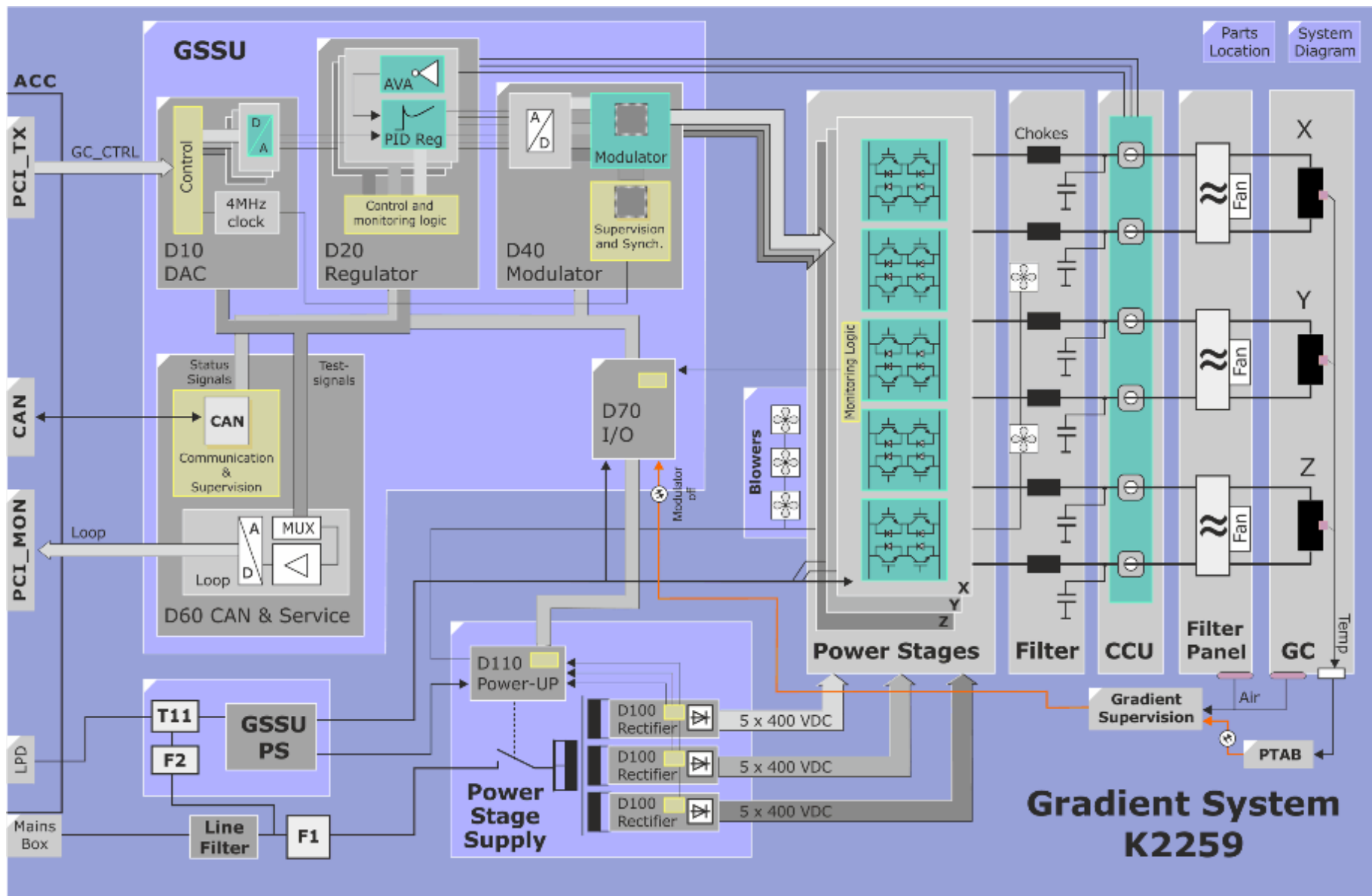
1.3.8 Gradient Coil Assembly

The Gradient Coil assembly consists of the active-shielded water-cooled Gradient Coil and five integrated high-order shim coils for active shimming and pockets for shim trays for placing shim iron plates for the passive shim. All connections are at the rear side of the magnet.

Tab. 2 Gradient Coil Type

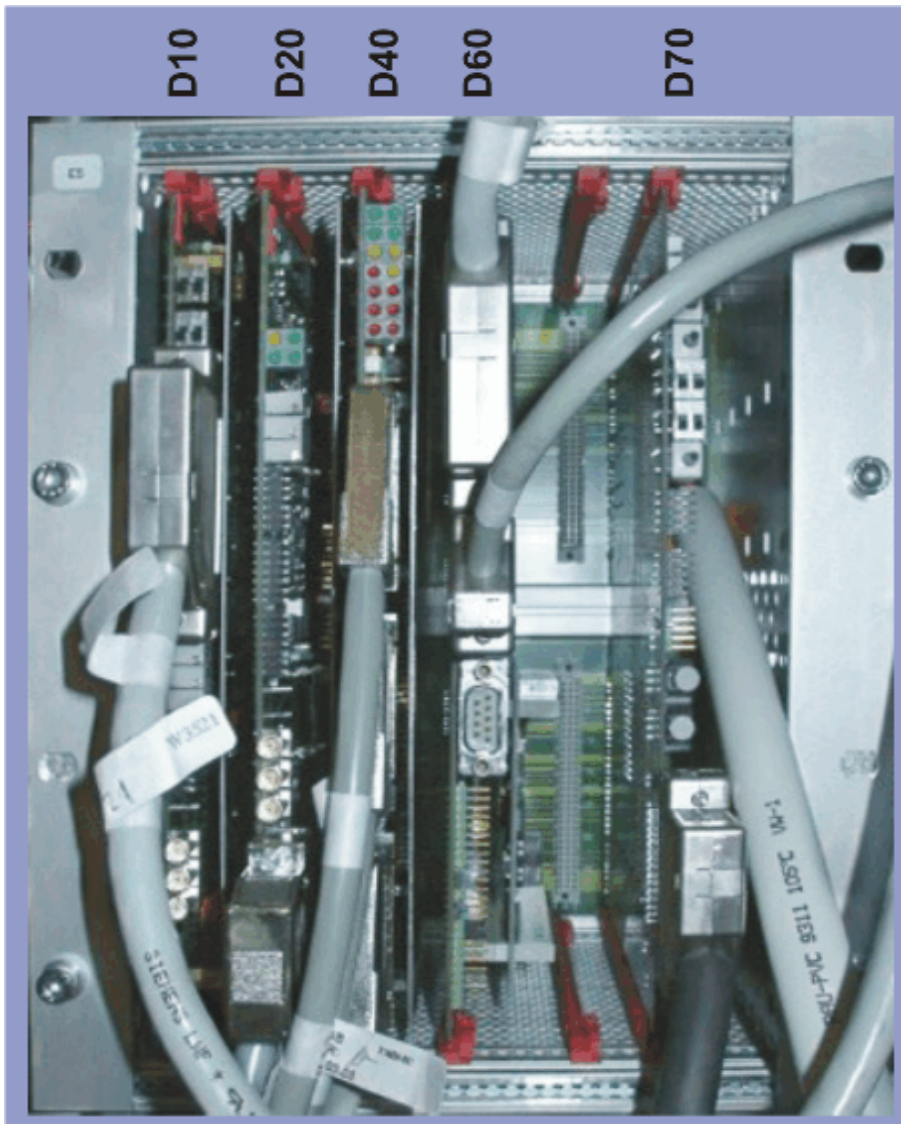
Coil	Description
AS 092	Trio Gradient Coil. Shim trays are accessed from the FRONT of the coil.

Fig. 2: Block Diagram of the Gradient System K2259



2.1 Overview

Fig. 3: GSSU Front View



The GSSU is responsible for generating the pulse-width modulated control signals for the switches in the power stage, based on the digital data supplied by the AMC. Its additional task is the overall supervision of the GPA.

The GSSU consists of the following PCBs or FRUs (Field Replaceable Units):

- Assembly Carrier
- DAC D10
- Regulator D20
- Modulator D40
- Service/CAN D60
- I/O board D70

2.2 Assembly Carrier

2.2.1 Function

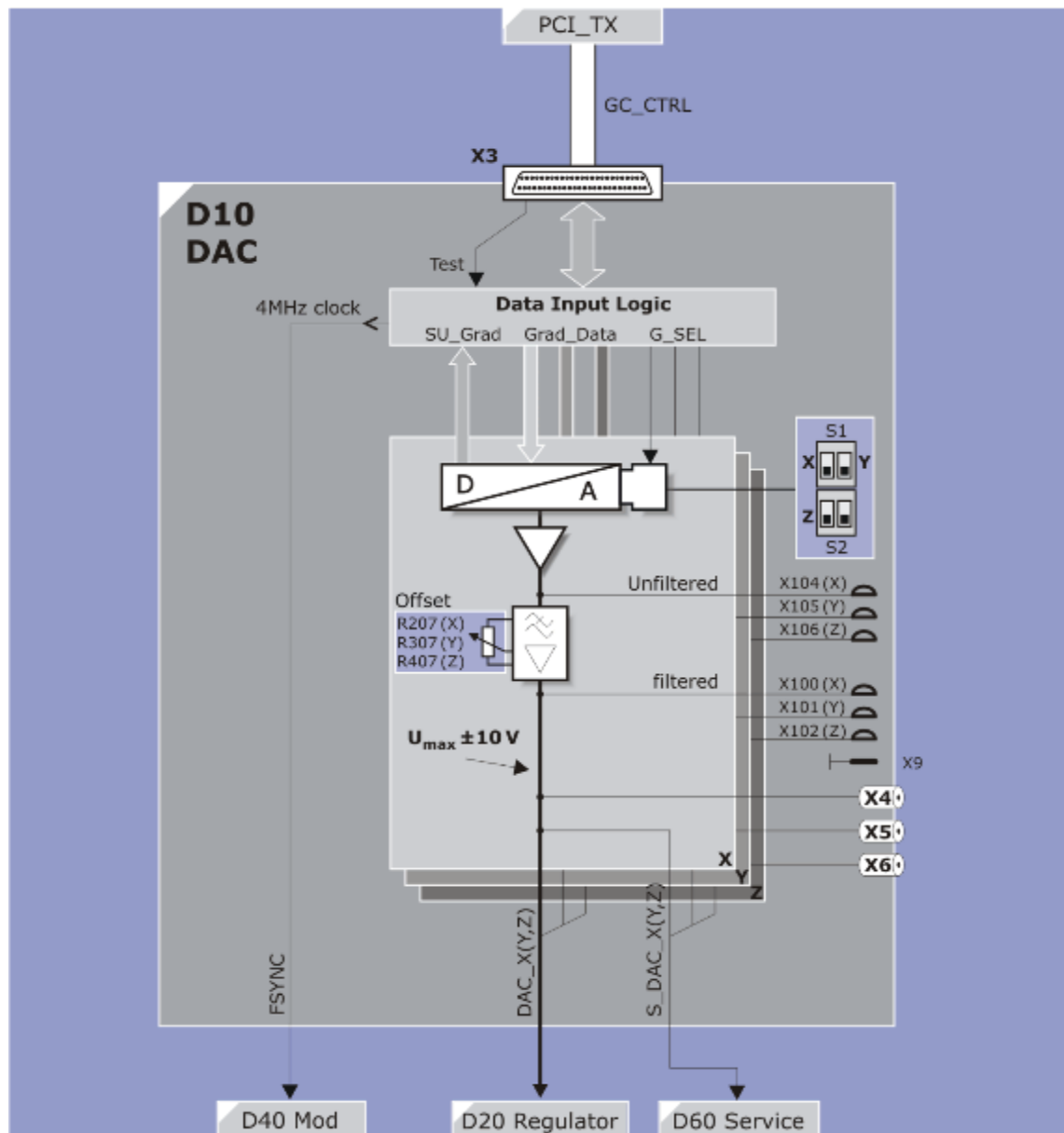
The Assembly Carrier consists of the mounting rack for the GSSU boards and backplane D80 providing interconnection between the GSSU components. A fan is built in on top of the Assembly Carrier.



The complete Assembly Carrier is a Field Replaceable Unit (FRU).

2.3 D10 DAC

Fig. 4: D10 DAC



2.3.1 Function

The DAC-board contains the D/A converters for all three X, Y, and Z channels. The digital data supplied by the AMC contains the gradient pulse amplitude data including the eddy current pre-distortion and the gradient delay. After the data are converted into analog signals, they are sent to Regulator D20 as the nominal current value. Additionally, a 4 MHz clock signal (from AMC) is sent to Modulator D40 where it is used for synchronization of the gradient.

2.3.1.1 Control Logic

The 20 bit amplitude data (**GRAD_D**) for the three gradient axes is transferred multiplexed. The select signals **G_SEL** are used to clock in new gradient data values.

2.3.1.2 Input Loopback Test

An input data loopback test between the PCI_TX (in the AMC) and the D10 can be run from the Service Software. When the **Test** signal is activated (SeSo), the 20 bit data of the X channel (**SU_Grad**) and the control lines can be looped back to the PCI_TX to verify the data path.

2.3.1.3 Nominal Value

The main DAC output is fed over the GSSU backplane to the D20 Regulator board as the nominal gradient amplitude. A copy is also sent to the D60 CAN & Service board where it is used as a feed back to the measurement control for loop testing in the Service Software (DAC test).

The output voltage of the DACs (**DAC_X, Y, Z**) is ± 10 V at full scale. It corresponds to the maximum gradient current that is configured by the GPA Software license.

2.3.1.4 Synchronization

A 4 MHz clock signal is also delivered from the AMC for use in synchronizing the GPA to the rest of the MR system.

2.3.2 Switches

The switches **S1** (left/right) and **S2** (left) can be used to apply digital zero to the input of the DACs. This is necessary for offset adjustments using potentiometers R207(X), R307(Y) and R407(Z).

Tab. 3 DIP switches S1, S2

Switch	Description	Position
S1 left	DAC_X enable/disable	down/up
S1 right	DAC_Y enable/disable	down/up
S2 left	DAC_Z enable/disable	down/up
S2 right	not used	not used

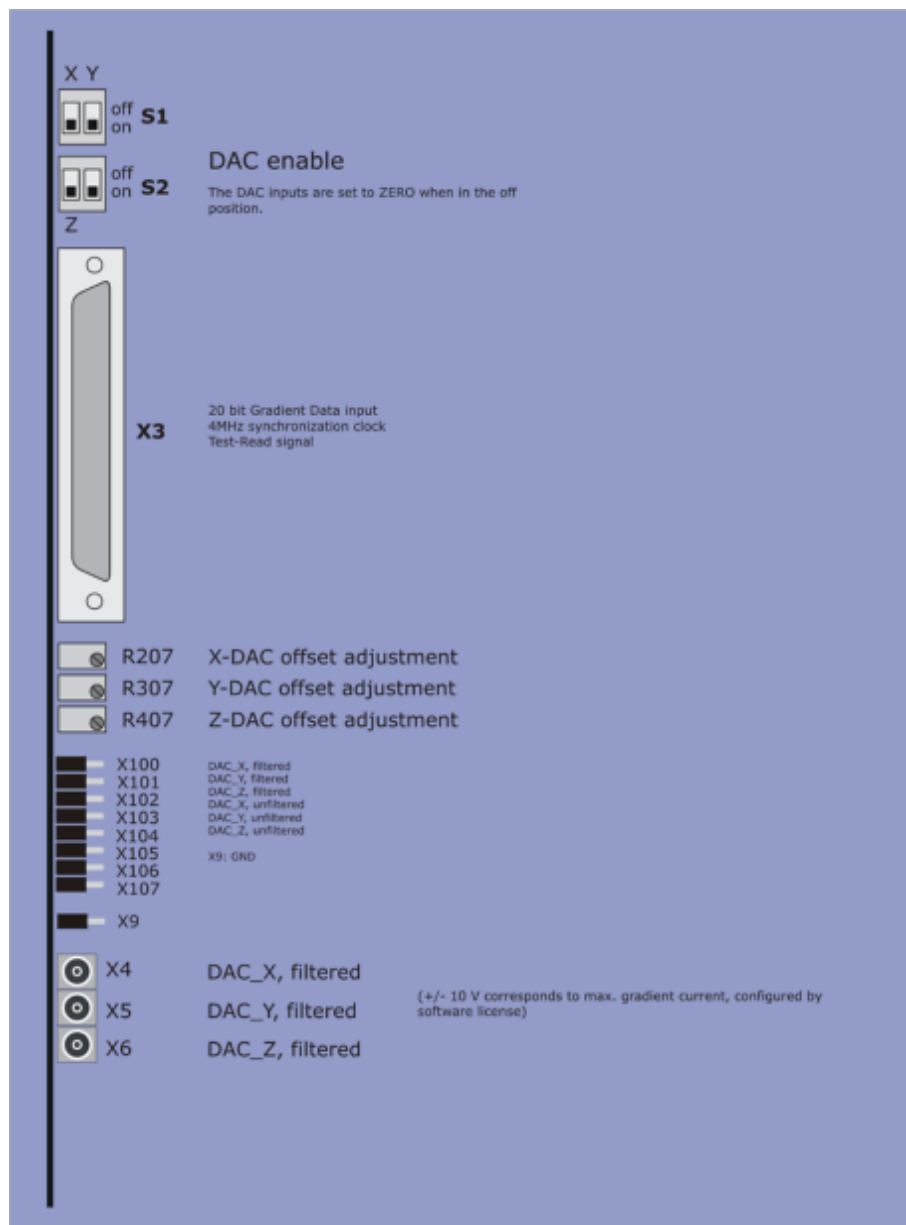
2.3.3 Test Points

The DAC output signals (DAC_X, Y, Z) are available for monitoring via QLA connectors (X4, X5 and X6). The test pins X100-X107 and X9 are intended for factory tests.

2.3.4 Potentiometers

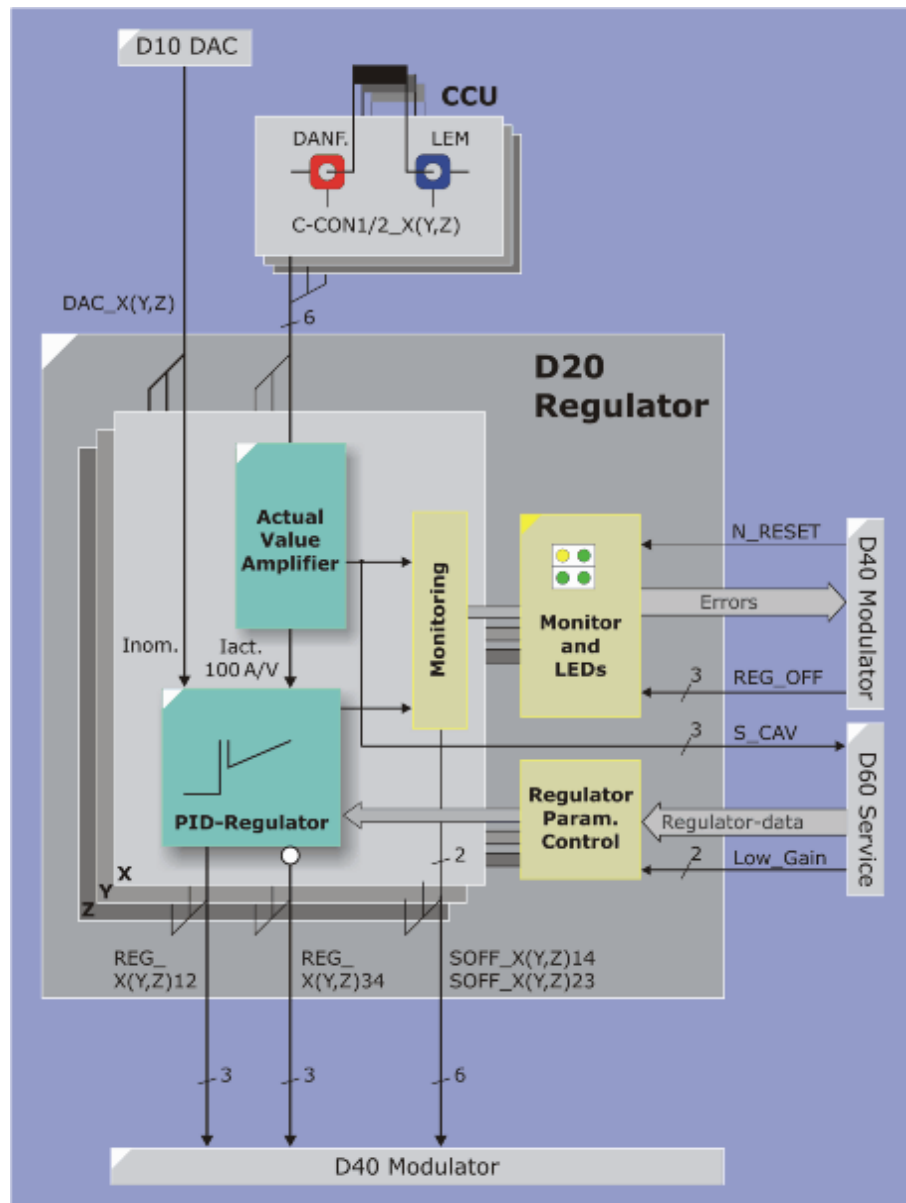
The static DAC or cabinet -offset can be adjusted to zero with the help of three potentiometers. During the adjustment, the corresponding DAC has to be disabled by using DIP switches S1 and S2.

Fig. 5: D10 Front View



2.4 D20 Regulator

Fig. 6: D20 Regulator Block Diagram



2.4.1 Overview

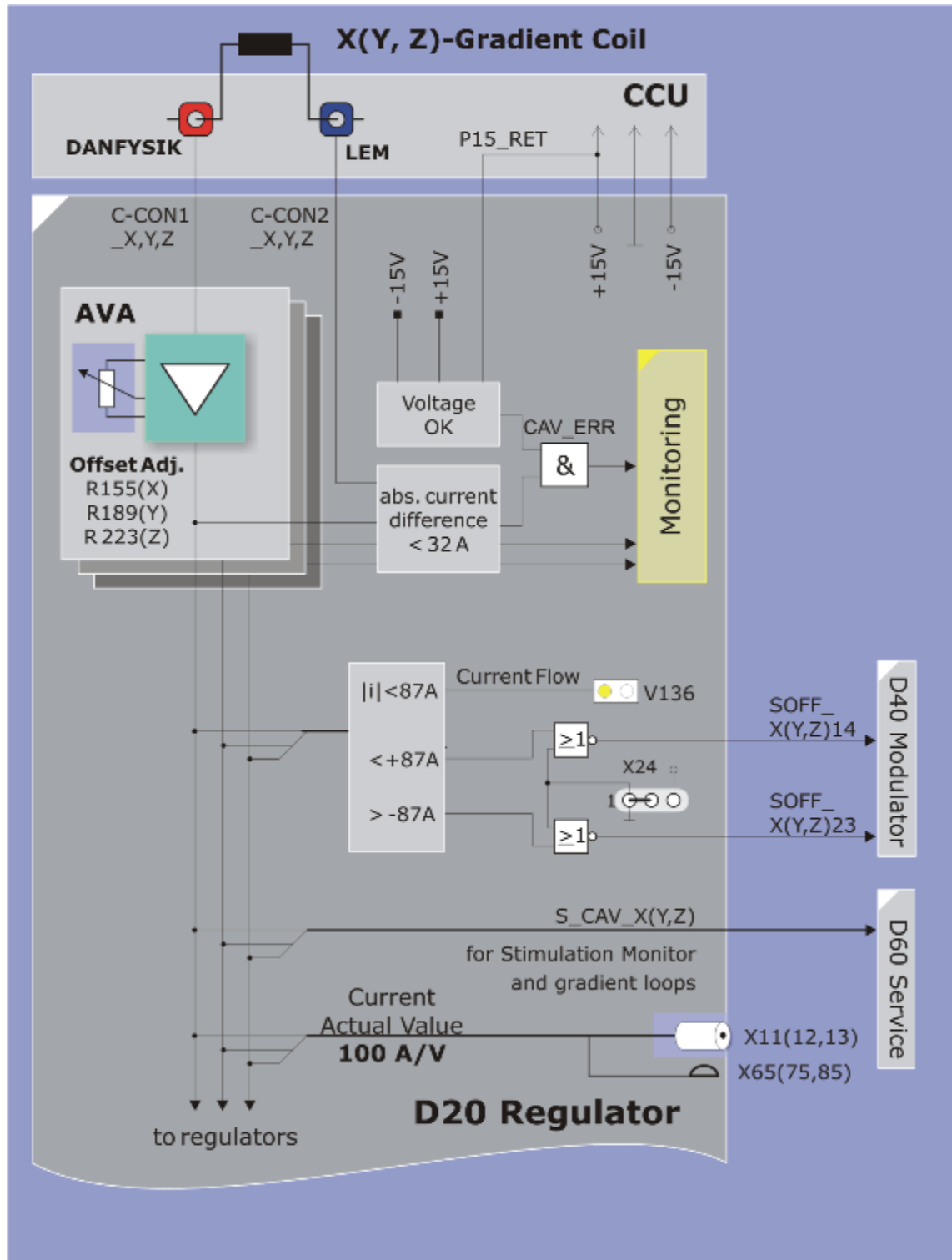
The D20 includes the Regulator function for all three axes. It contains the following functions:

- Actual Value Amplifier
- PID Regulator
- Regulator Parameter Control
- Monitoring and LEDs

2.4.2 Function

2.4.2.1 Actual Value Amplifier (AVA)

Fig. 7: D20 Actual Value Amplifier



The DANFYSIK current sensors in the Current Converter Unit (CCU) deliver a current (C-CON1_X(Y,Z)) that is proportional to the current driven through the Gradient Coil. That current is converted to a corresponding voltage level (1 V corresponds to 100 A in the coil).

The amplifier has **offset** adjustment potentiometers R155(X), R189(Y) and R223(Z) for amplifier offset compensation. Switches S3/4 on the D70 are used to ZERO the current for making this measurement.

The **C-CON2_X(Y,Z)** signal from the LEM current sensor is only used for monitoring functions and is not amplified.

Stimulation Monitoring and Gradient Loops

The AVA's main output signal is used for regulation. A copy (**S_CAV_X(Y, Z)**) is sent to the D60 CAN & Service board where it is digitized and sent to the PCI_MON in the AMC for peripheral nerve stimulation monitoring. In case the slew rate exceeds critical values, the PCI_MON switches off the gradient current (see D40 and D60 description). The same signal is also used to evaluate the gradients during Tune-up and service test procedures (gradient loops).

The current actual value (or I_{act}) is available at the front side over QLA connector X11(12,13) where it is used, for example, to adjust the AVA amplifier offsets.

Monitoring and LEDs

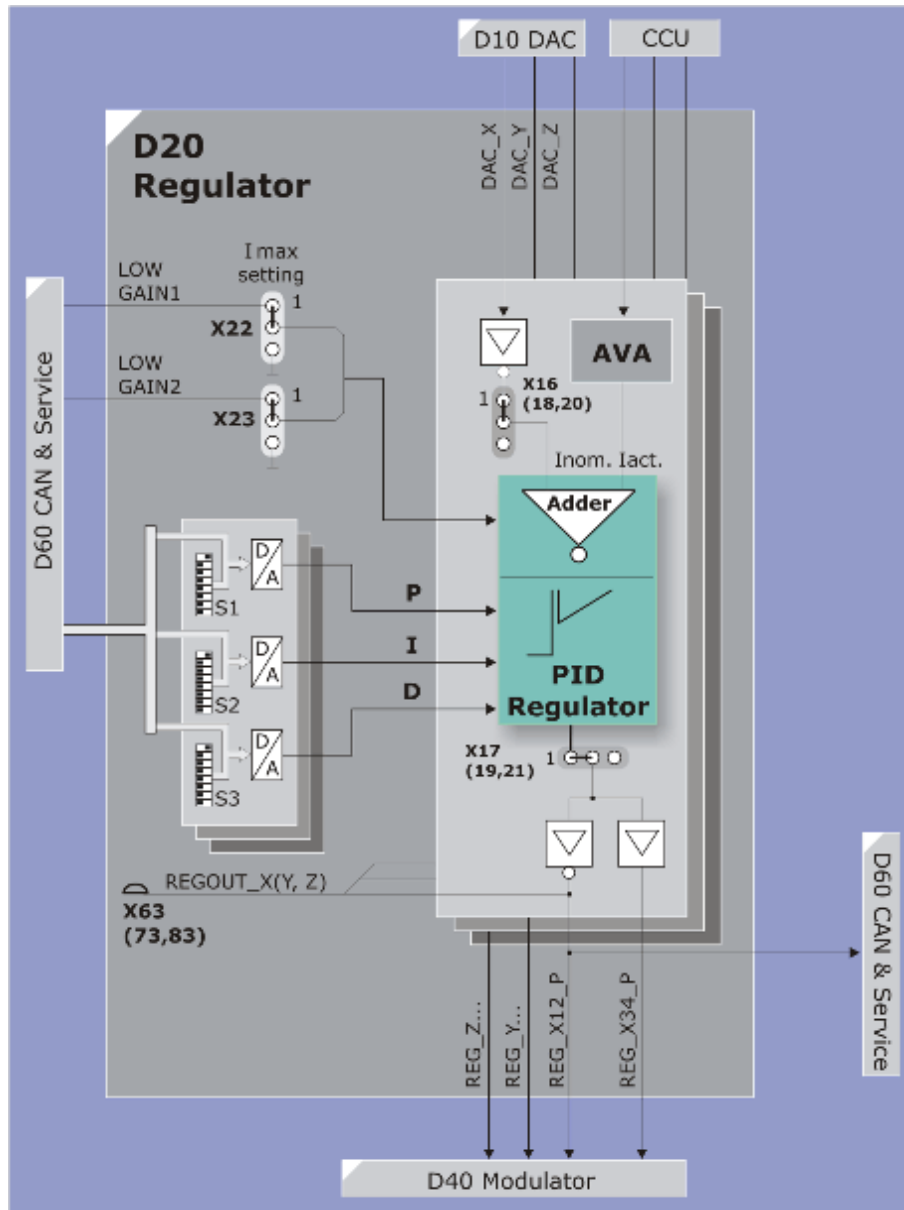
The DANFYSIK and LEM outputs are compared, if the difference is less than 32 A, there is no error.

The power for the DANFYSIK and LEM devices (± 15 V and ANA_GND) is supplied by the D20-board and the +15 V is looped back as P15_RET-signal for monitoring. The ± 15 V supplies are also monitored.

If any of the above monitored signals indicates an error, the respective current actual value error **CAV_ERR_X(Y, Z)** is sent to the D40-board to disable the Modulator (for description of other errors and LEDs see D20 monitoring description).

2.4.2.2 PID-Regulator

Fig. 8: D20 PID-Regulator



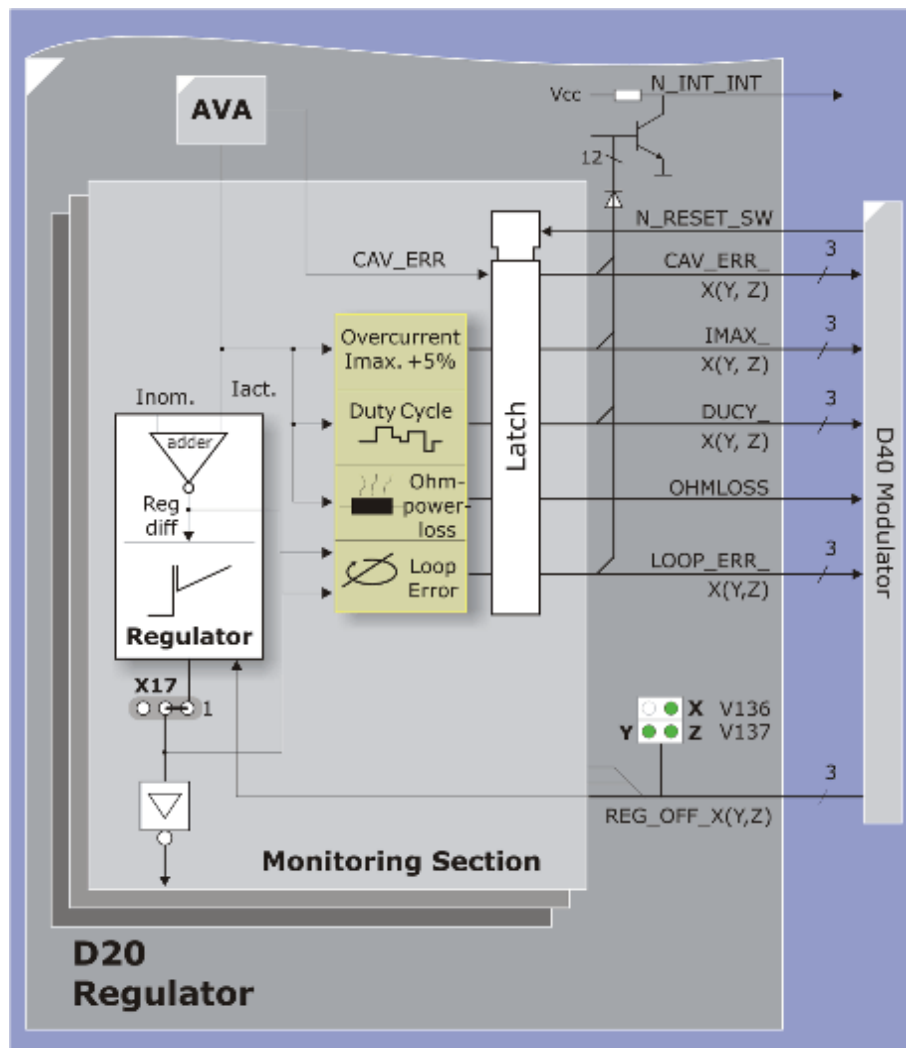
The nominal value sent from the DAC-board is inverted and added to the actual value coming from the AVA. The signal is fed into a closed loop regulator circuit, based on a digitally controlled 7-bit PID-Regulator (P=proportional, I=integral, D=differential part). The regulator output signals are sent to the Modulator board D40 (inverted and non-inverted).

Regulator Parameter Control

The initial configuration of the Regulator after power on is determined by DIP-switches S1 to S9 (see description of switches). The configuration of switch 8 (off-position) on switch S3, S6 and S9 determines if this initial hardware setting can be overwritten by Software via the D60. In that case, the downloaded site-specific Regulator parameters from the Tune-up are activated.

2.4.2.3 Monitoring

Fig. 9: D20 Monitoring



The following table shows the monitoring on the D20 board:

Tab. 4 D20 monitoring

Error	Monitored signal
CAV-error	current sensor difference >32 A; 15 V return from CSA; ±15 V supply

Error	Monitored signal
Overcurrent	actual current value
Duty cycle	actual current value
Ohm power loss	actual current value
Regulation (loop) error	Regulator difference, regulator output signal

All of the above mentioned errors are reported to the D40 Modulator board. From there, the D60 CAN & Service board are informed about the error condition.

Regulator Enable

The Modulator board D40 uses the signal **REG_OFF_X(Y, Z)** to disable/enable the regulator board. The enable status is shown by three green LEDs on the front panel of the D20 board.

Disable Logic

The actual value is monitored for a threshold of 87 A absolute. If the current is higher, the signal **SOFF_X(Y, Z)14** disables the positive current direction and the signal **SOFF_X(Y, Z)23** disables the negative current direction at the Modulator board D40 as indicated by LED V136 left (lights up).

2.4.3 Switches

Tab. 5 D20 DIP-switches (regulator settings)

Axis	P	I	D
X	S1	S2	S3
Y	S4	S5	S6
Z	S7	S8	S9

2.4.4 LEDs

Tab. 6 D20 LEDs

LED	Description
V136 left	current flow (one or more axes) > 87 A
V136 right	Regulator X enabled

LED	Description
V137 left	Regulator Y enabled
V137 right	Regulator Z enabled

2.4.5 Test Points

Refer to sticker inside gradient cabinet.

2.4.6 Potentiometers

Tab. 7 D20 potentiometers

Potentiometer	Description
R155	X-current sensor offset adjustment
R189	Y-current sensor offset adjustment
R223	Z-current sensor offset adjustment

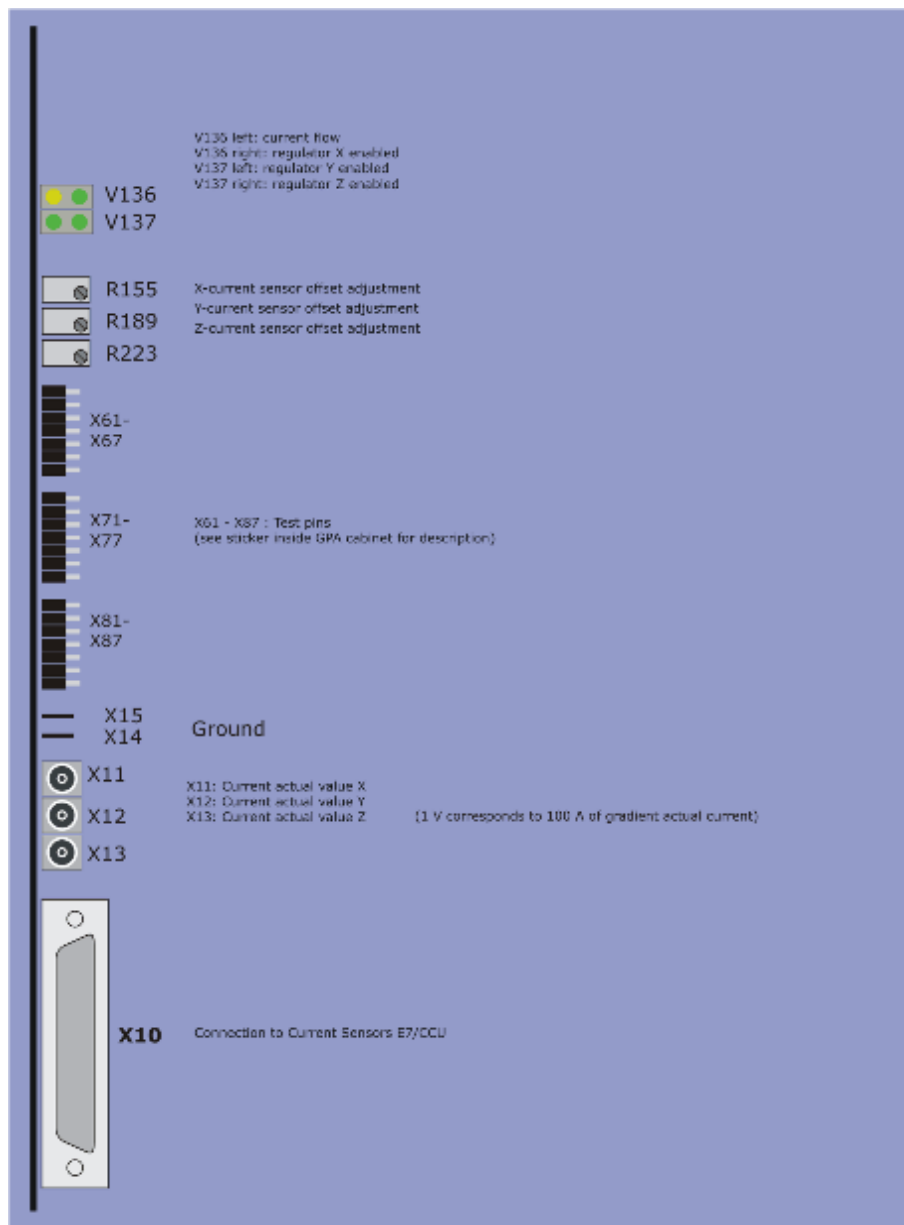
2.4.7 Jumpers

Tab. 8 D20 jumpers

Jumper	Position	Description
X22/23	1-2/ 1-2	D60 determines current scaling: TQ = 650 A
X24	1-2	Opposite current direction on the D40 Modulator is blocked by the disable logic

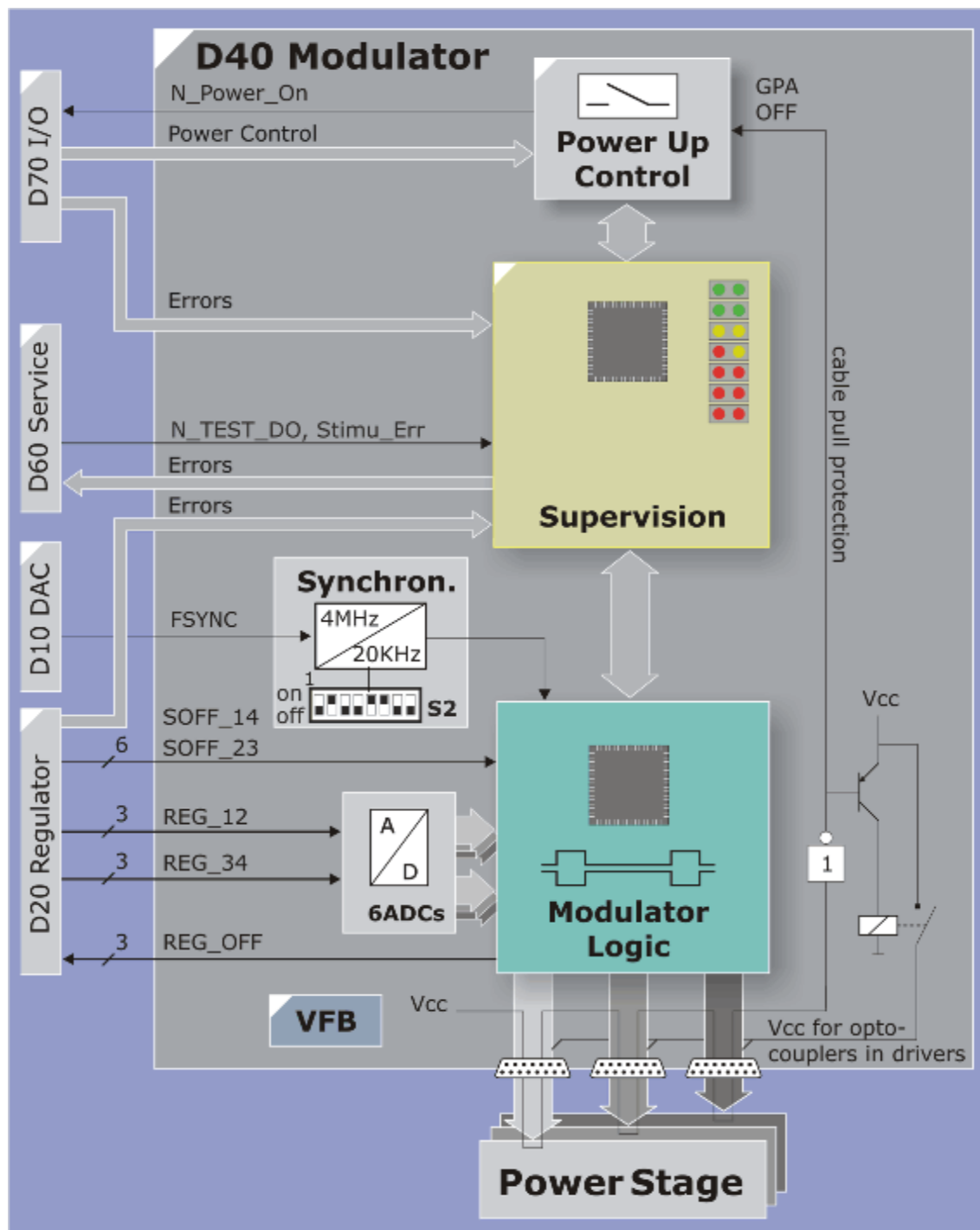
Jumpers X16-21 have to be set to position 1-2 for normal operation.

Fig. 10: D20 Front View



2.5 D40 Modulator

Fig. 11: D40 Modulator



2.5.1 Function

- **Modulation** - generation of pulse-width-modulated switching signals for the three axes.
- **Supervision** - circuit of time-critical power-stage related signals.

- Synchronization
- Power-UP control

2.5.1.1 Pulse-Width Modulator

The D40 board is based on a programmable logic. The D20 Regulator output signals REG_X(Y, Z)_12(34)_P(N) are analog-to-digital converted and fed to the Modulator chip where pulse width modulation takes place. The switching frequency is 20 kHz. It is synchronized by a clock signal that is derived from the 4 MHz FSYNC by a programmable frequency divider. The switch signals for the power stages are fed out by three front plugs.

By means of a standard SCSI-cable (it is not a SCSI-bus!), the signals are connected to the power stage modules. The proper connection of those is monitored by a wire bridge.

The D20 Regulator is able to report errors to the D40 Modulator. In case of an error, the D40 can disable the corresponding Regulator axis by means of signal REG_OFF_X(Y, Z).

2.5.1.2 Slew Rate Adjustment

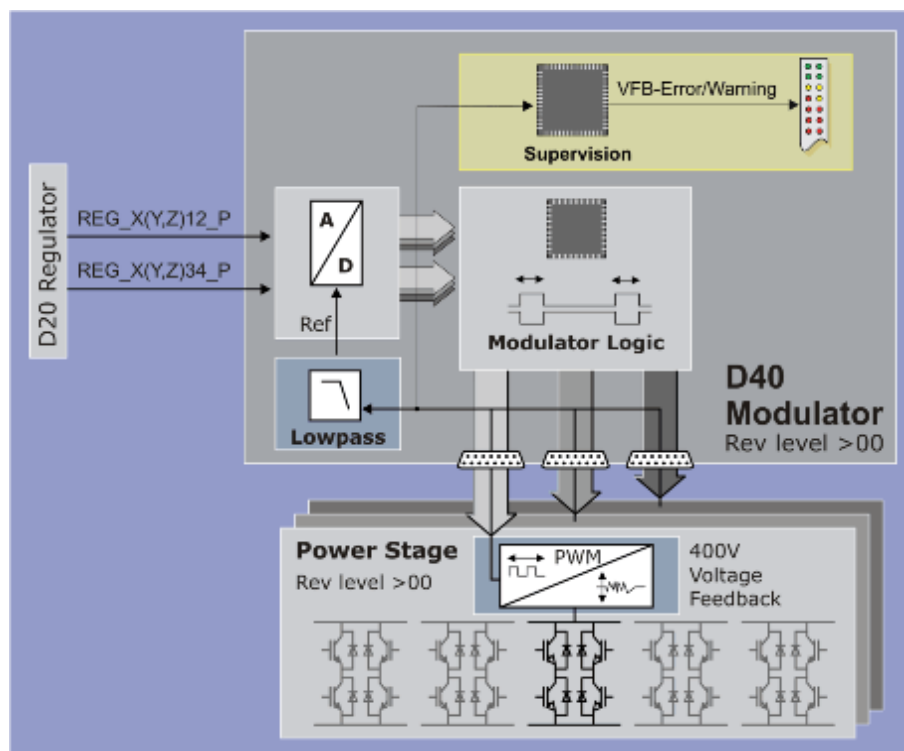
The maximum possible slew rate of the Gradient System is adjustable by setting limits for the pulse-width Modulator. The configuration of jumpers X10 and X11 determines if Software limits from the D60 or fixed values from the board itself are used (see jumper list).

2.5.1.3 Power Stage Voltage Feedback (VFB)

Gradient Power Stages and D40 Modulator boards with "revision level > 00" are equipped with a final stage supply voltage feedback circuit that controls the pulse width of the Modulator chip. By these means, the negative effects of a fluctuating 400 VDC supply voltage on the gradient current stability are largely eliminated. The compensation works for PS voltage levels from 250-450 VDC. Outside that range or if the feedback voltage is faulty, an axis-dependent voltage feedback (VFB) error is generated.

A D40 Modulator board with "rev. level > 00" cannot be combined with GPA power stages showing a "Rev. level = 00" since the feedback circuit would be missing. All other combinations will work properly.

Fig. 12: D40 Modulator VFB



2.5.1.4 Supervision, Display, Synchronization

Voltage Monitoring

The D40 has a built-in internal monitoring for its supply voltages (± 15 V and $+5$ V = VCC).

Modulator Logic Monitoring

- **Oscillation:** Regulation loop oscillation with more than 10.5 kHz
- **Clock fault:** The Modulator logic monitors the presence of the 164 MHz internal clock.
- **Wrong controlling signal:** The proper timing and logic status of the power stage control signals is continuously checked (e.g. the two transistors of a half-bridge must not be switched on at the same time). In case of a problem, the Modulator is switched off.
- **Modulator defect:** The "clock fault", "wrong controlling signal" and some other D40-internal supervisions result in a "Modulator defect" error, indicated by the red LED (V123, left).
- **Cable pull protection:** A wire bridge between pin 1 and pin 50 inside the "SCSI"-plug in the power stage modules is used to check the proper connection of all plugs. The 5 V (VCC) control signal is looped via all connectors. When interrupted, the VCC-supply for the opto-transmitters in the power stage's driver boards is switched off and the "Cable pull protection" error is generated.
- **Power Stage cable defect:** During GPA Power-UP, the Modulator logic checks the functions of the output stages and the cable connection to the power stages by switching the output shortly (0.5 sec.) to "high impedance." The monitoring at the SU-VI will measure "high level", but in case of a faulty Modulator output or a broken wire

"low level" across the 20 k Ω pull-down resistor. A "Power Stage cable defect" error is generated as a result.

Fig. 13: Power Stage Cable Test

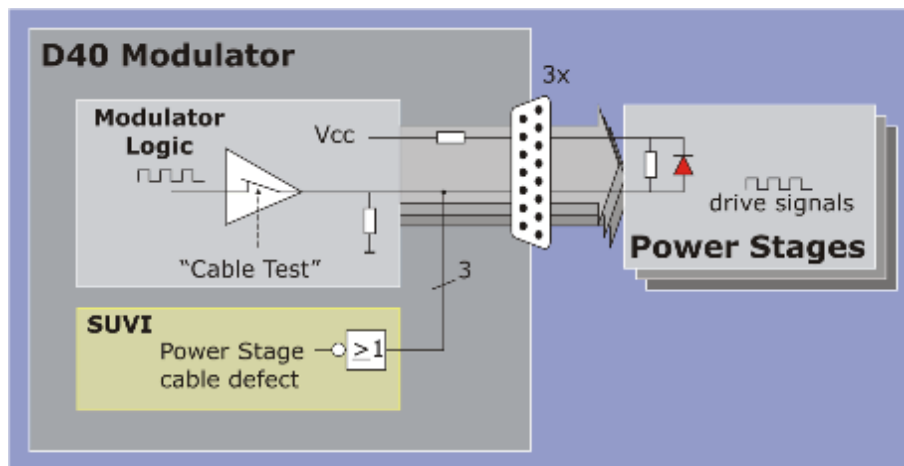
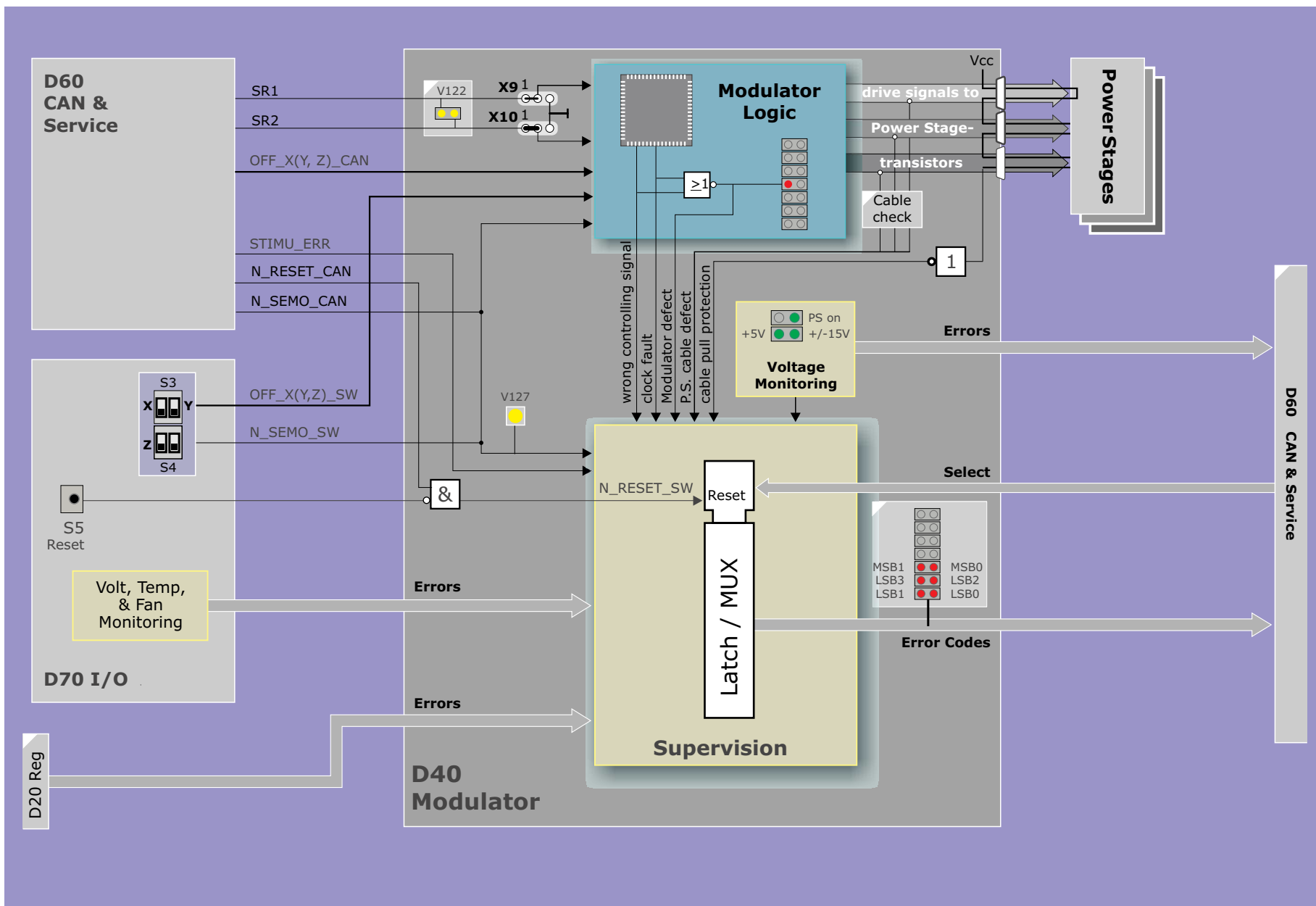


Fig. 14: D40 Supervision, Display, Synchronization



SUVI

The supervision logic (SUVI) reports the errors generated by the Modulator logic and handles the following additional errors:

- **Synchronization fault:** 20 kHz sync. not ok.
- **All Regulator errors:** reported by the D20 Regulator board.
- GPA and Gradient Coil **overtemperature**, GPA **voltage** and **fan** monitoring: reported by several monitor circuits via the D70 I/O board.
- **Stimulation errors:** reported by the Software Stimulation Monitor (PCI_MON in the AMC)

The SUVI error control logic can report 64 different error codes. These are addressed by the D60 board with Select lines. The error code is read out and displayed by 6 red LEDs at the front side of the board (see LED description). A warning or error message is generated and fed to the D60 board.

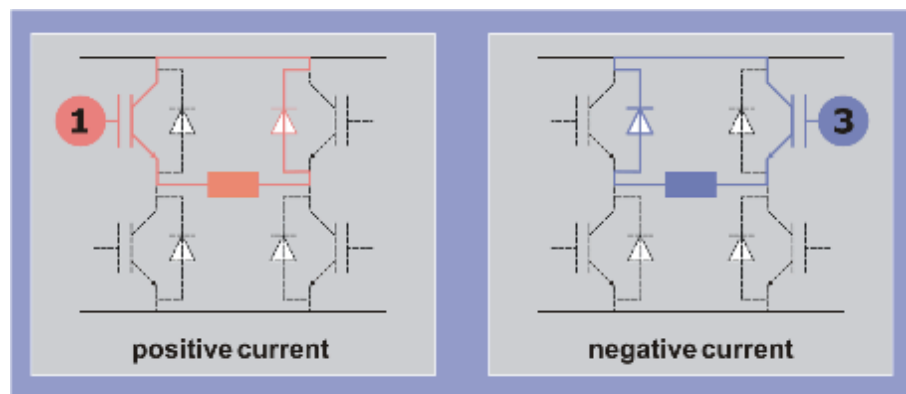
The SUVI can generate three different types of fault classes which will be treated in greater detail in the LED description.

Switch-Off Modes

In case of an error, the Modulator can disable a running gradient current in three ways:

- **"Hard" stop:** All switches of the power stage are blocked (e.g. VCC error, wrong controlling signal error...).
- **"Soft" stop:** the Modulator permanently enables switch 1 and 3 of the power stage, i.e. upper free-wheeling (e.g. I_{max.}, DUCY, synchronization error, coil temp. error, stimulation error...)

Fig. 15: Soft Stop



- **Power-off** the power stage supply (e.g. power stage supply over/under voltage...)

Modulator Disable

The Modulators can be disabled with service switches S3, S4 at the D70 (**OFF_X(Y,Z)_SW**) or by Software **OFF_X(Y, Z)_CAN** signal from D60 board). Both cases will result in a "hard switch-off" of the gradient current.

Reset of Errors

All errors can be reset by the processor D60 (**N_RESET_CAN**) or manually by push button **S5** at the I/O-board D70.

Service Mode

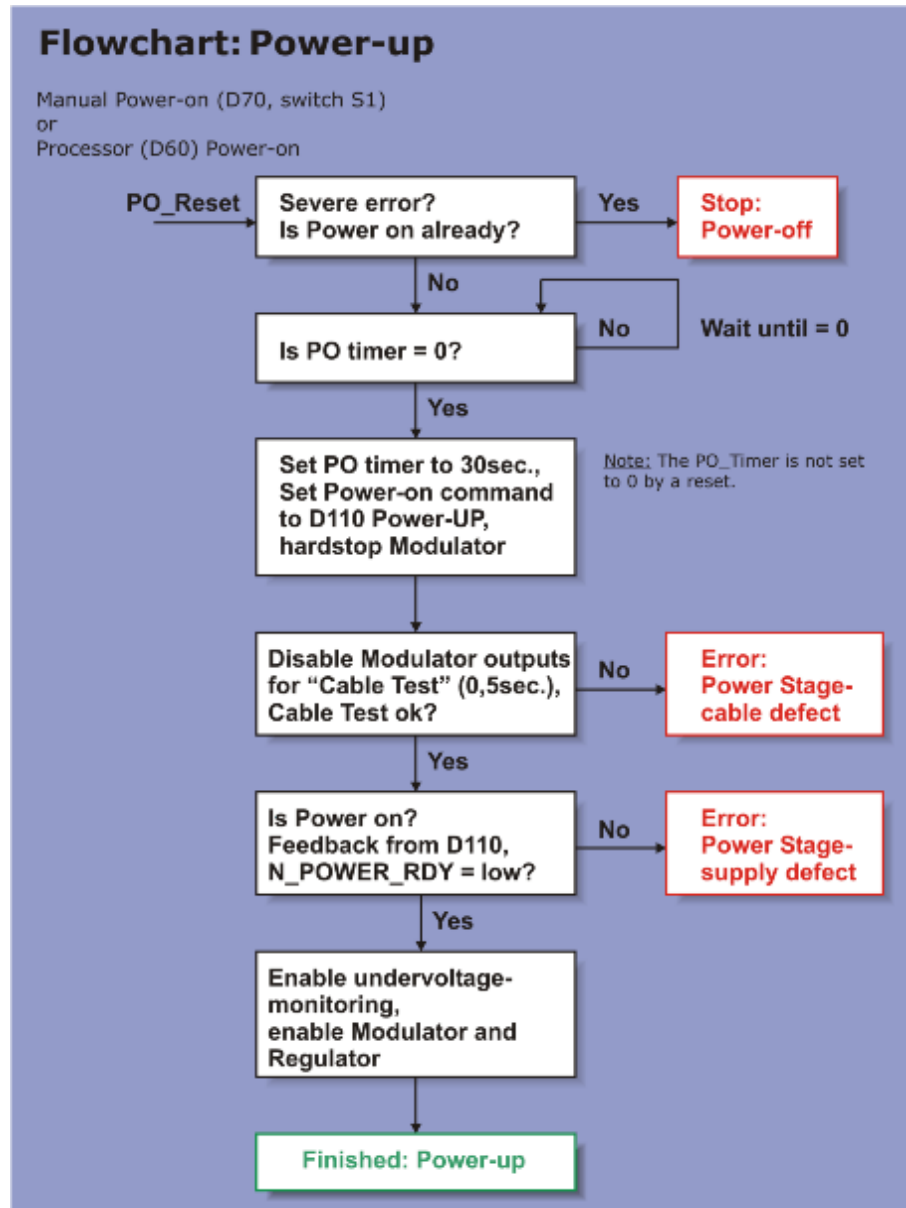
To facilitate troubleshooting, the GPA can be switched to Service Mode by a switch (S4 right at D70 in upward position) or by Software (**N_SEMO_CAN** signal from D60 CAN & Service board).

The following functions are affected by the Service Mode:

- **Modulator enable:** The Modulators are not disabled when the power stage supply voltage is switched off.
- **Synchronization fault:** The fault is handled as a "warning" only.
- **Closed loop error:** The fault is handled as a "warning" only.
- **"Soft" stop:** No Modulator "soft" stop in case of low voltage faults at the power stage.
- **Voltage feedback error:** The fault is handled as a "warning" only.
- **Oscillation error:** Regulator may oscillate for a longer period of time (approx. 12 sec. instead of 3 sec.) until the error is reported.

2.5.1.5 GPA Power-UP

Fig. 16: GPA Power-up Flowchart



The D40 controls the power stage supply switching-on and -off (see also D110 description). At the beginning of the Power-UP routine, all stored errors are reset, if an error occurs later on, the D40 will power-off the supply and generate the corresponding error message.



Power-UP can take up to 30 seconds due to an internal timer at the D40 Modulator-board.

2.5.2 Switches

Tab. 9 D40 DIP Switches

Switch	S1	S2	S3	S4 (D40 Rev. >04)
1	off	off	off	off
2	off	ON	off	ON
3	off	off	off	ON
4	off	off	off	off
5	off	ON	off	n.a.
6	off	ON	off	n.a.
7	off	off	off	n.a.
8	off	off	off	n.a.



D40 boards from spare parts stock are delivered with default jumper setting which leads to the error "Modulator defect"! Correct the jumper setting before use!

2.5.3 LEDs

16 different error codes can be displayed in dual code with help of four error LEDs (V125 - V126 = LSB 3..0). Furthermore, the logic can identify two different fault classes (indicated by LED V124 left/ right = MSB1/ 0):

- common faults for all axes
- axis-specific faults for X, Y and Z-axis

The first severe error that is recognized is always displayed as an LED pattern that lights up very bright, subsequent errors are displayed in a darker pattern.

Tab. 10 D40 LEDs

LED	Status	Description
V120 left:	on flashing	all Modulators on all Modulators on, but warning
V120 right	on flashing	Power stage power supply on Power status is changing
V121 left	on	+5V ok.
V121 right	on	+/-15V ok.

LED	Status	Description
V122 left/ right		max. output voltage 500V max. voltage 1000V max. voltage 1500V max. voltage 2000V (TQ-gradient)
V123 left	on	Modulator defect
V123 right:	on	Warning
V124 left/ right		see next pages
V125/ V126		see next pages
V127	on	Service Mode on



A minor problem is indicated as “warning” only (yellow LED V123/ right lights up additionally).

Fig. 17: D40 Front View

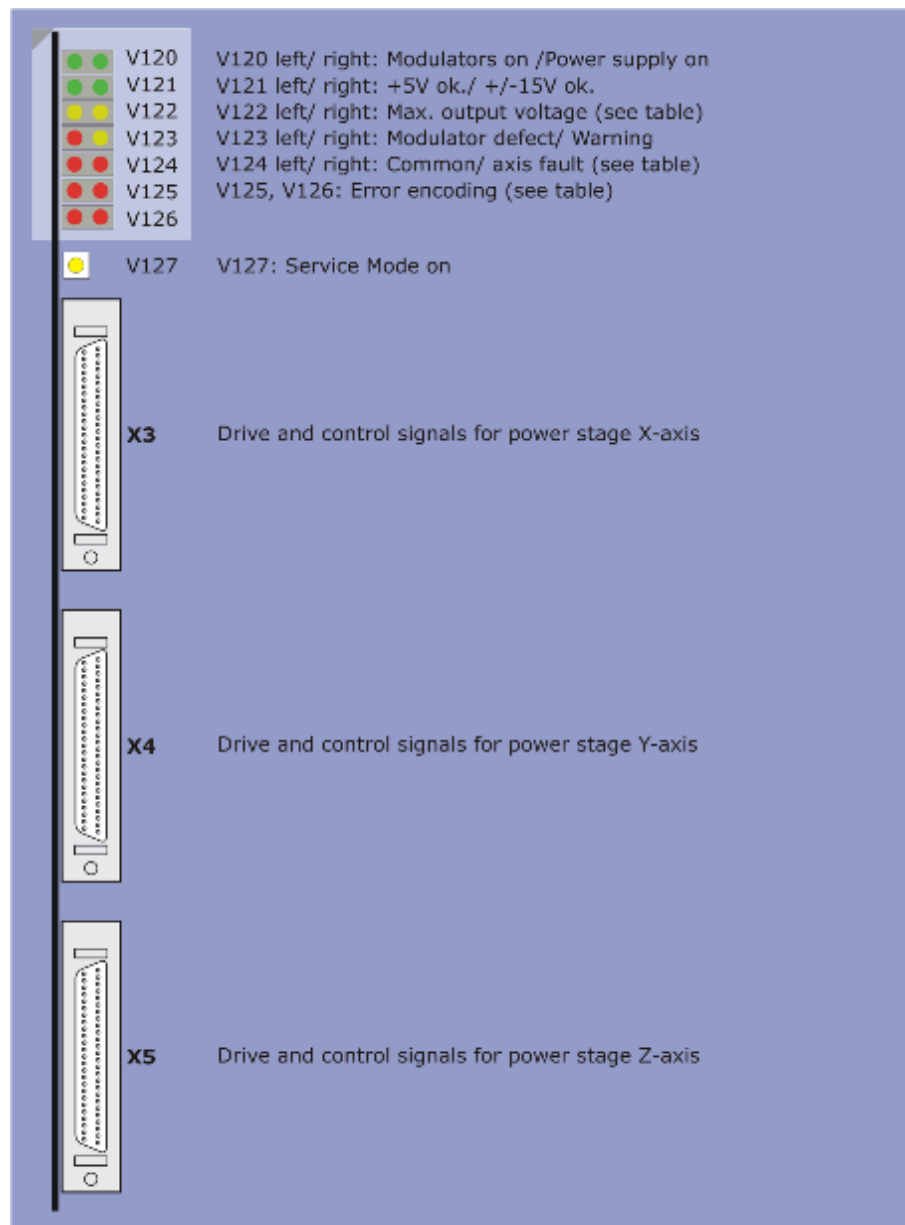


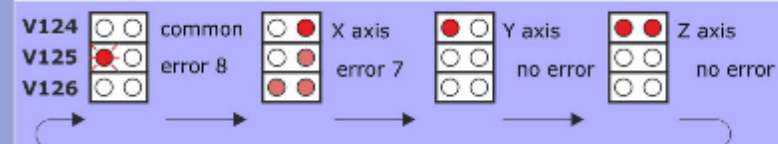
Fig. 18: D40 LED Error Patterns

Error	LED V125/126	LED V124	Description
0	LEDs flash 4 times	common	Oscillation
		X, Y, Z	Voltage feedback error
1		common	GPA error; look for other errors
		X, Y, Z	OFF by D60
2		common	Soft stop by D60
		X, Y, Z	switched off by D70 (S3, S4)
3		common	Fan-error
		X, Y, Z	Over current (Imax. > 688 A)
4		common	modulator: wrong controlling signal
		X, Y, Z	Duty cycle too high (>400 A _{DC})
5		common	cable pull protection
		X, Y, Z	Ohm power loss too high (>400 A _{eff})
6		common	switched off by Stimulation Monitor
		X, Y, Z	current value (sensor) error
7		common	not used
		X, Y, Z	Regulation (loop) error
8		common	Gradient coil temp. too high or Gradient Supervision Alarm
		X, Y, Z	control signal freq. too high
9		common	Power Stage defect (Power on error; extreme power)
		X, Y, Z	Driver power supply defect
10		common	GPA air temperature too high: >42°C warning - LED V123R on >47°C error
		X, Y, Z	Power Stage temperature too high

Error	LED V125/126	LED V124	Description
11		common	Modulator off (power is off)
		X, Y, Z	Power Stage over voltage
12		common	Power Stage supply defect
		X, Y, Z	Power Stage low voltage
13		common	Main Transformer temperature
		X, Y, Z	Rectifier temperature too high
14		common	Synchronization fault
		X, Y, Z	inductor (choke) temp. too high
15		common	open coil switch (not used)
		X, Y, Z	Power Stage cable defect

LED V124 :

common X axis Y axis Z axis

An example :

In the first LED pattern V124 indicates a common fault is being displayed and V125,126 show the error number 8 pattern. Also, this LED lights up very bright indicating this error occurred first.

Then in succession the X, Y and Z axis errors are displayed, whereby only the X axis is displaying an additional error, error number 7. Here the LEDs light up only at half brightness indicating it is not the first error that occurred, but a secondary error.

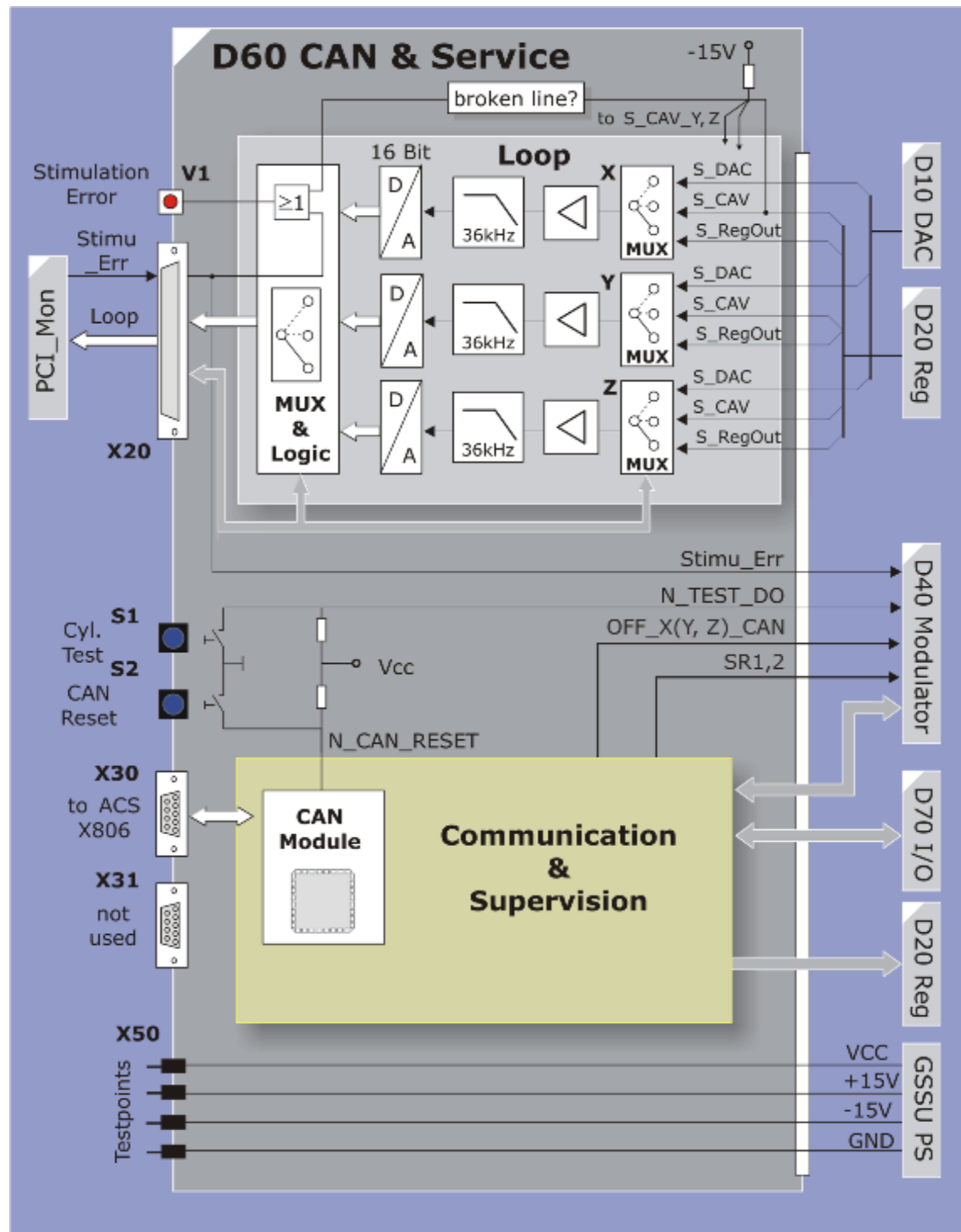
2.5.4 Jumpers

Tab. 11 D40 jumpers

Jumper	Position	Description
X6/ X7/ X8	open	not used
X9	1-2	Stimulation monitor is active
X10/ X11	1-2/ 1-2	Slew rate adjusted by D60
X12	open	not used, for PCB test only
X13, X14	1-2	normal operation/ PCB test

2.6 D60 CAN & Service

Fig. 19: D60 CAN & Service



2.6.1 Overview

The CAN & Service board consists of two functional blocks:

- **CAN module** (Communication and supervision)
- **Signal multiplexer** (Loop)

The CAN module is responsible for data transmission between the AMC and the GPA. Control data for the GPA is transferred from the AMC and GPA-status information to the AMC.

The signal multiplexer is used to connect signals generated in the GPA to the AMC (PCI_MON board). This link allows Software evaluation of the selected signals for stimulation monitoring, Tune-up and tests.

2.6.2 Function

2.6.2.1 CAN Module

Communication

The firmware (FW) stored in a PROM allows the processor to run a self-test and to communicate and load the application software over the CAN-Bus. The application software (LW) for the processor is stored in an EEPROM. According to this program, the processor is able to receive control data from the AMC and transfer status information from the GPA to the AMC.

Reset

The front panel push button S2 is used to reset and re-start (warm start) the processor.

2.6.2.2 Signal Multiplexer (Loop)

The signal multiplexer can select 3 analog GSSU signals from each gradient axis via the backplane:

- **Current actual value** S_CAV_X(Y, Z)
- **Nominal value** S_DAC_X(Y, Z)
- **Regulator output** S_REGOUT_X(Y, Z)

Tests and Tune-up

These signals are converted into digital data and fed to the AMC for Software evaluation, e.g. for tests and Tune-up under SeSo.

Stimulation Monitoring

During imaging sequences, the 3 current actual values $S_CAV_X(Y, Z)$ are converted into digital data and fed multiplexed via X20 to the PCI_MON board in the AMC for stimulation monitoring. In case of exceeding critical thresholds, the PCI_MON sets the Stimu_err signal and the D60 will "soft stop" the Modulator.

2.6.3 Switches

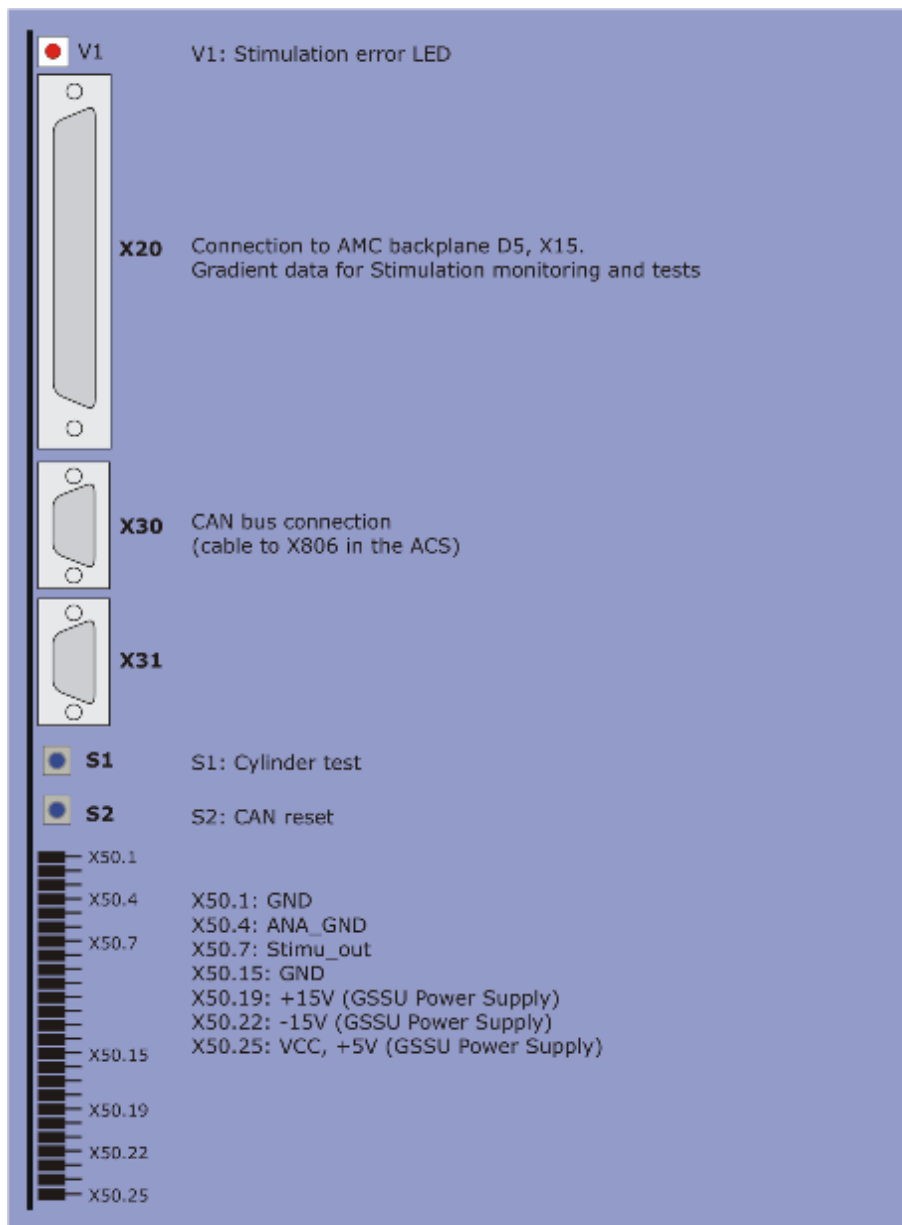
Tab. 12 D60 switches

Switch	Description
S1 (push button)	Cylinder test (not used)
S2 (push button)	performs reset of the CAN controller

2.6.4 LEDs

V1 (stimulation error) lights up if the critical gradient stimulation values are exceeded.

Fig. 20: D60 Front View

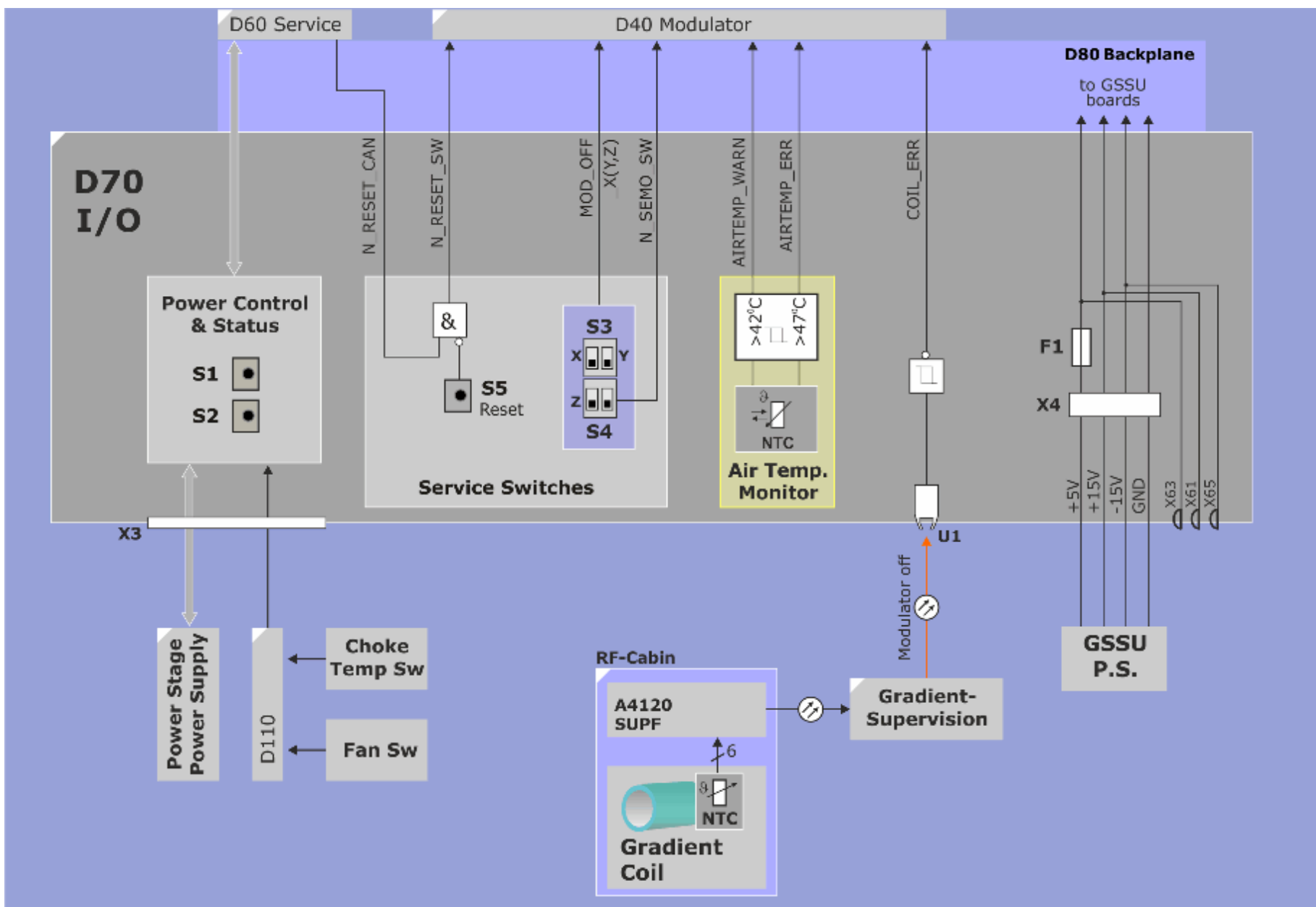


2.7 D70 I/O

2.7.1 Overview

The I/O board is an interface board between the GSSU and the GSSU power supplies but also to the power stage section. Control and monitoring signals are fed via this board to the Modulator board D40 and the D60 CAN & Service board.

Fig. 21: D70 I/O Overview



2.7.2 Function

2.7.2.1 Power Switching

The D40 Modulator board controls the power switching that can be initiated by processor D60. For service functions, the GPA can be switched on or off by push buttons S1 and S2 on the D70 board. See D110 description for details.

2.7.2.2 Voltage Supply for the GSSU boards

The +5 V (VCC) and the +/-15 V from the GSSU power supply are supplied by connector X4 of the D70. Buffer capacitors for all voltages and a 6.3 A fuse (F1) for the +5 V supply are located there. The buffered voltages are supplied via the backplane to all GSSU boards and can be measured at test points at the front side of the board.

2.7.2.3 Gradient Temperature Monitoring

The following temperature-related signals are monitored via the D70 I/O board:

- **GPA air temperature** - an on-board NTC monitors for air over temperature. The AIR-TEMP_WARN (>42 °C) and AIRTEMP_ERR (>47 °C) signals inform the D40 Modulator board about that problem.
- **Gradient Supervision Alarm** - in case the Gradient Supervision has detected smoke particles (e.g. from overheated gradient connections), the light at FOC D70/U1 will be removed and the GPA will be disabled with Modulator off. In addition, a relay at the VLM opens and via the AMC a warning pop-up is posted to the console (Scanner Hardware Error. Automatic Shutdown! Evacuate patient...).

One minute later, the VLM opens a second relay contact and powers-off the MR-system to stand-by.

- **Gradient Coil Temperature** - the temperature of the Gradient Coil is measured with six thermal resistors (NTCs), moulded inside the Gradient Coil and connected via X50 to the monitoring circuit SUPF (Supplied Functions, A4120-board) at the Patient Table Control Electronics frame. From there, the signal is routed via fiber optic connector to the Gradient Supervision and forwarded to FOC D70/U1.

In case of gradient coil over temperature, the light at FOC D70/U1 will be also removed and the GPA will be disabled with modulator off. The GPA will inform the AMC and a warning pops-up at the console (Gradient Power Amplifier Error: "Temperature of the gradient coil is too high").

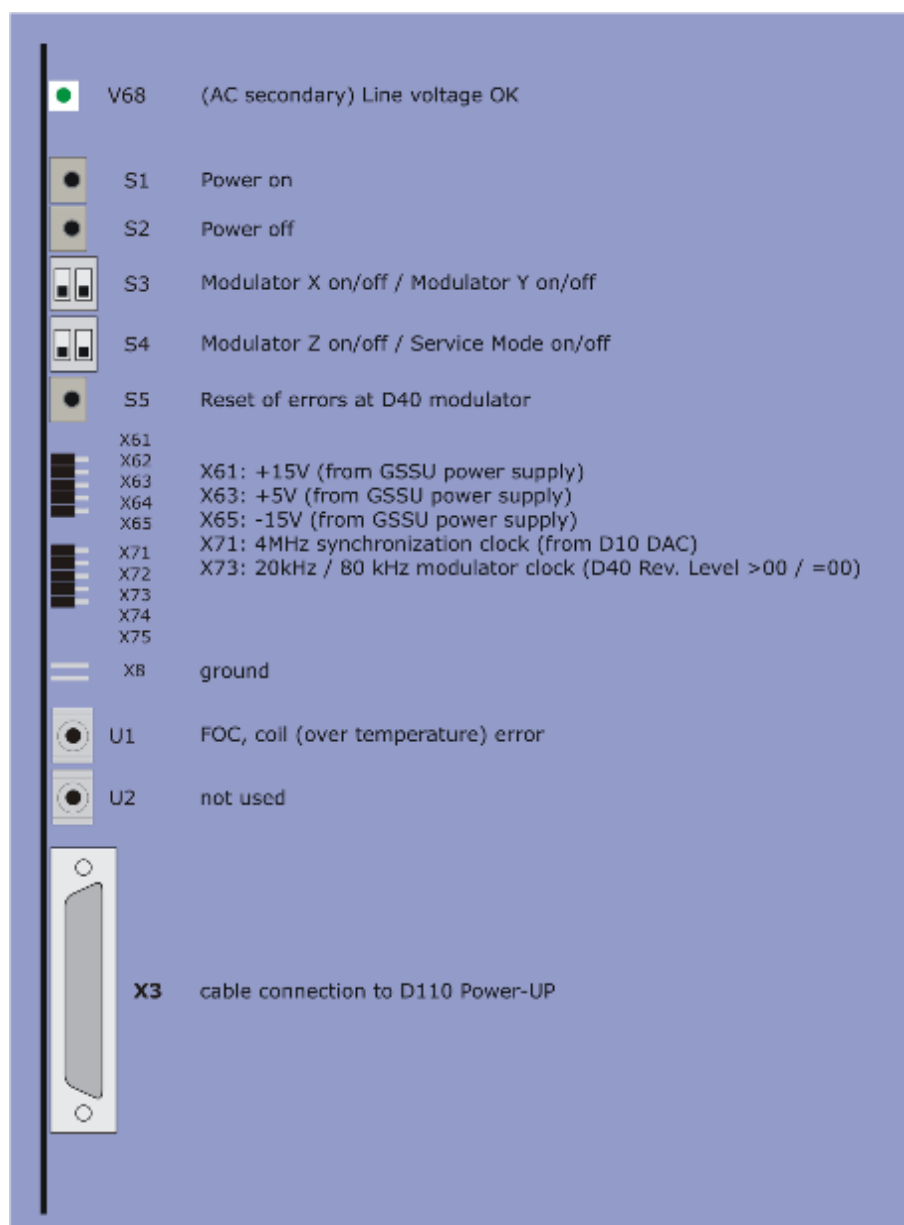
2.7.3 Switches

Tab. 13 D70 switches

Switch	Description	Position
S1 (push button)	Power-on switch (mains power)	
S2 (push button)	Power-off switch (mains power)	

Switch	Description	Position
S3 left	Modulator X on/off	down/up
S3 right	Modulator Y on/off	down/up
S4 left	Modulator Z on/off	down/up
S4 right	Service Mode off/on	down/up
S5 (push button)	Reset of errors (at D40)	

Fig. 22: D70 Front View



2.7.4 LEDs

One green LED V68 at the front side of the D70 board indicates LINE_VOLT ok (secondary voltage = 283 VAC, tolerance +/-10 %).

2.7.5 Fuses

There is a 6.3 A fast-blow fuse F1 on board for the +5 V voltage.

2.7.6 Test Points

The +5 V (VCC) and the +/-15 V from the GSSU Power Supply can be measured at test points on the front side of the D70 board as well as two important timing signals: 4 MHz synchronization clock and the 20 kHz / 80 kHz (depending on D40 board revision level) Modulator internal clock.

2.7.7 Jumpers

By means of inserting jumpers X15 and X16, each of the errors passing the D70 board can be jumpered out. In this case, the desired error is not monitored anymore!



CAUTION

This is a troubleshooting feature only.

For normal operation the X15 and X16 jumpers must be set according to the jumper list!

» Otherwise, severe damage to the GPA and the Gradient Coil may occur !

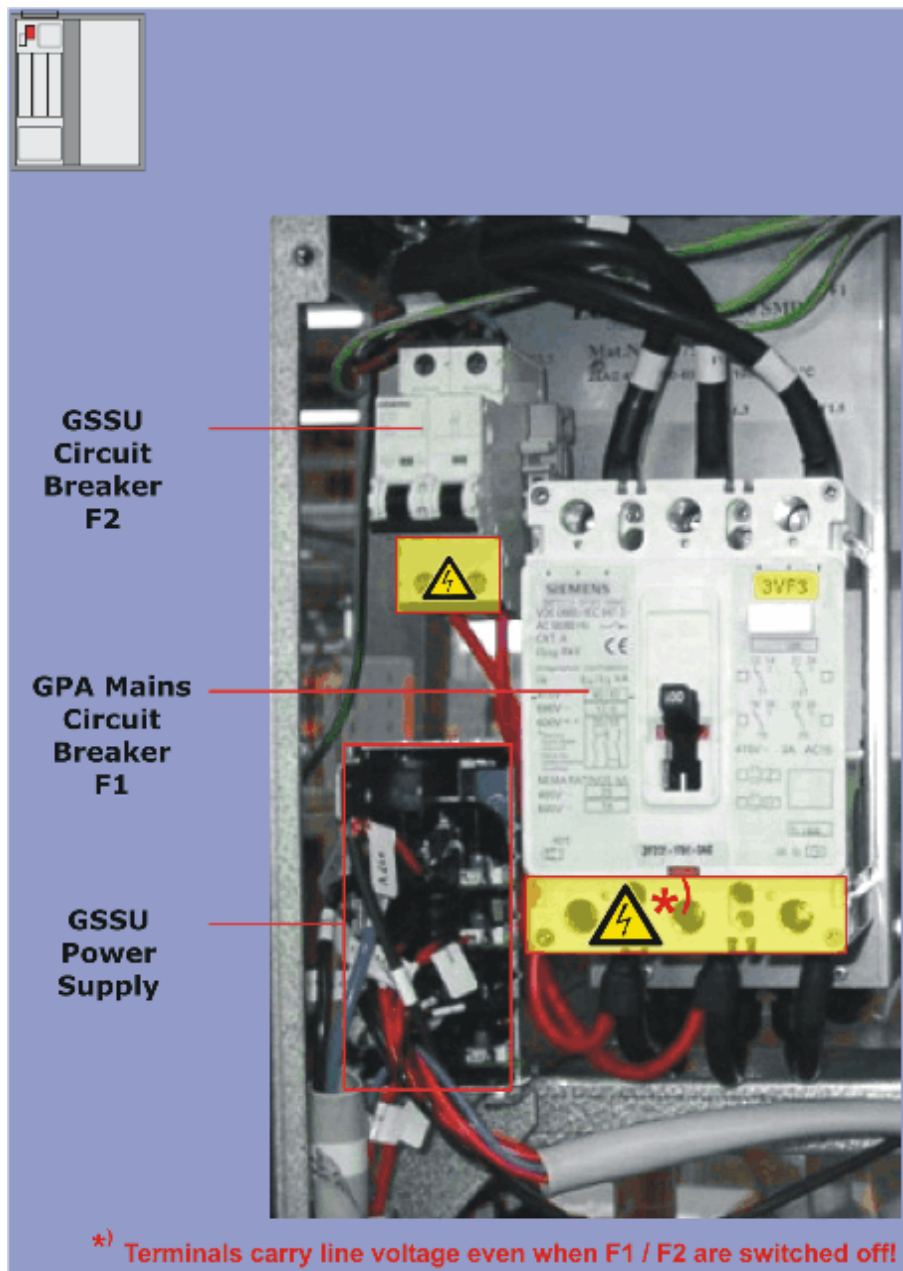
Tab. 14 D70 jumpers

Jumper	Position	Description
X15	A1-B1	Simulates N_POWER_READY
X15	A3-B3	Bypasses POWER_ON_ERR
X15	A5-B5	Bypasses FAN_ERR
X15	A9-B9	Bypasses RC_TEMP_X error
X15	A10-B10	Bypasses RC_TEMP_Y error
X15	A11-B11	Bypasses RC_TEMP_Z error
X15	A12-B12	Bypasses FI_TEMP_X error
X15	A13-B13	Bypasses FI_TEMP_Y error
X15	A14-B14	Bypasses FI_TEMP_Z error
X16	A1-B1	Bypasses TRAFO_TEMP error

Jumper	Position	Description
X16	A2-B2	Simulates LINE_VOLT ok.
X16	A4-B4	Bypasses the UV_X error
X16	A5-B5	Bypasses the UV_Y error
X16	A6-B6	Bypasses the UV_Z error
X16	A7-B7	Bypasses the OVW_X error
X16	A8-B8	Bypasses the OVW_Y error
X16	A9-B9	Bypasses the OVW_Z error
X16	A11-B11	Bypasses the COIL_ERR (coil temperature)
X16	A14-B14	Bypasses AIRTEMP_WARN/ _ERR

3.1 Overview

Fig. 23: GSSU Power Supply



CAUTION

The GSSU Power Supply is connected to mains by circuit breaker F2 directly! Switching-off F1 will not remove power from the GSSU.

Risk of electrical shock!

- » Switch off F2, before working at the GSSU power supply.

3.2 Function

The GSSU Power Supply generates three stabilized DC voltages for the GSSU boards, a 17 VDC voltage for the drivers in the Power Stage modules and a 24 VDC voltage for the relays and fans in the Power Stage supply section.

3.3 Test Points

All stabilized DC voltages are adjustable via potentiometers on the respective power supply.

Tab. 15 GSSU Power Supply

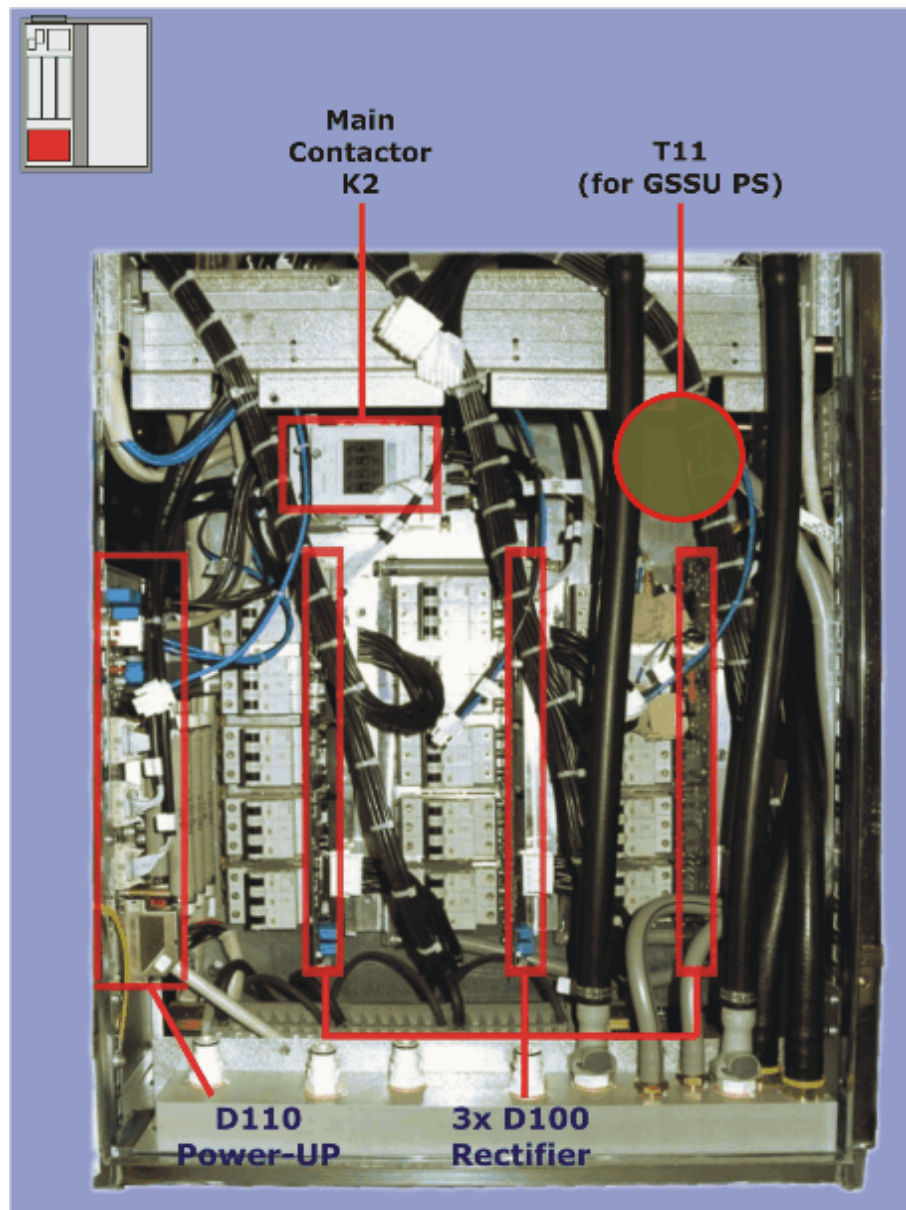
Slot #	Voltage	Test point
1	17.0 VDC $\pm$ 100 mV (driver supply)	Output connectors, Slot 1
2	5.075 VDC $\pm$ 25 mV (GSSU boards)	D70 X63/X8
3	+/-15.05 VDC $\pm$ 25 mV (GSSU boards)	D70 X61,X65/X8
4	24.0 VDC $\pm$ 500 mV (Power Stage relays and fans)	Output connectors, Slot 4



Voltages are measured with disabled Modulators (S3 left/ right and S4 left at D70 set to upward position).

4.1 Overview

Fig. 24: Power Stage Supply



CAUTION

The GPA is equipped with a dual power supply.

Risk of electric shock!

- » Switch off both circuit breakers F1 and F2 before starting to work at the GPA. Remember, there is still power at the line filter and circuit breaker input terminals!

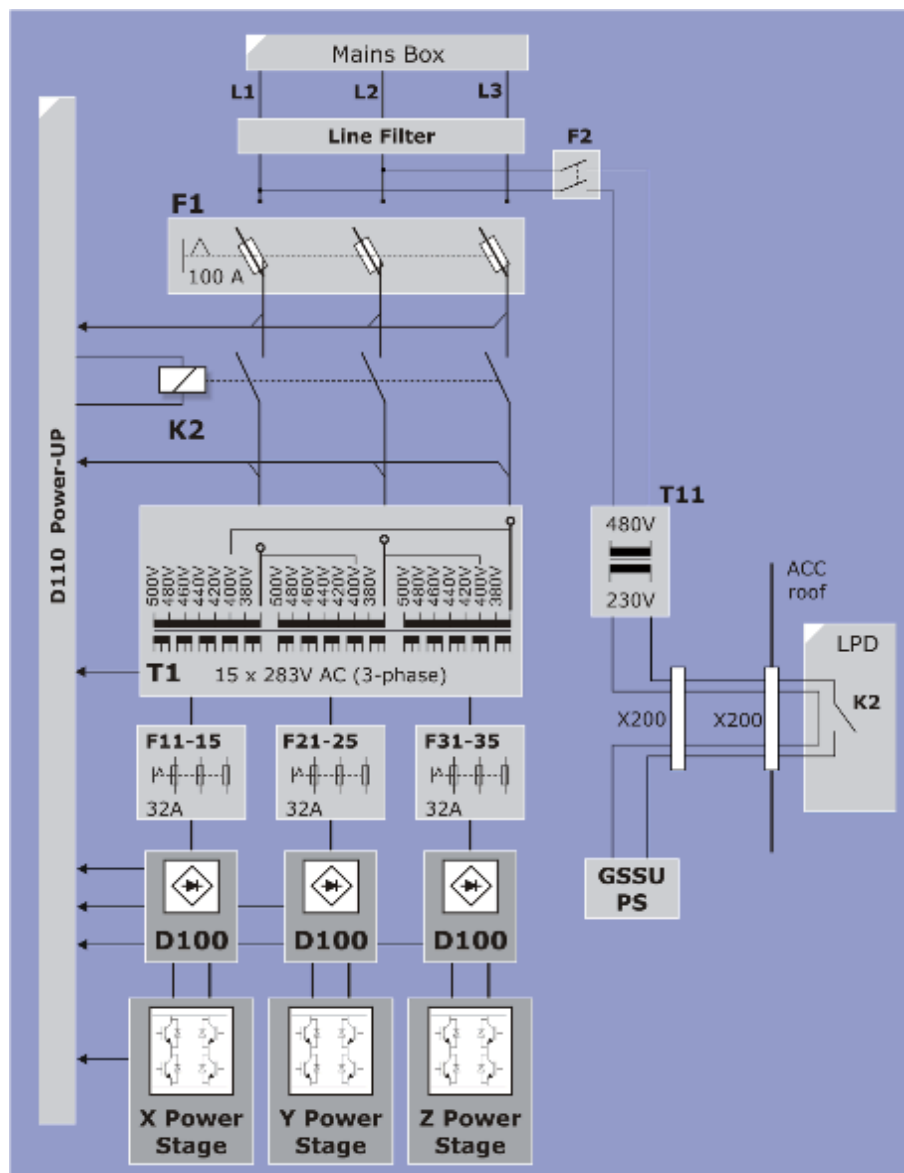
4.2 Overview

The mains voltage (L1, L2, L3) is taken directly from the customer mains and fed over a line filter and a 100 A circuit breaker F1 to transformer T1. Its primary tapplings need to be adjusted according to the local line voltage. The presence of the line voltage is shown by three yellow LEDs (V101, V201, V301) at the D110 Power-UP board.

To avoid extreme current peaks, the mains voltage is switched to transformer T1 via a "soft start unit" (D110 Power-UP). First, relay K1 energizes and connects the current limiting resistors to the transformer, then K2 activates and establishes a direct line connection. The T1 secondary provides fifteen 283 VAC 3-phase outputs for the Power Stage supply. The AC voltages are fed via fifteen 3-phase 32 A circuit breakers to the three D100 Rectifier boards. After rectification, each D100 provides five 400 VDC outputs for the cascades in the Power Stage.

4.3 D100 Rectifier

Fig. 25: Power Stage Supply



4.3.1 Function

4.3.1.1 Voltage Generation

Each of the D100 Rectifier boards is equipped with five 3-phase rectifiers that are supplied by the 283 VAC secondary outputs of the main transformer T1. The rectifiers are water-cooled by the cooling system. At the output of the rectifiers, five independent 400 VDC supply voltages for the Power Stages are available.

4.3.1.2 Rectifier Temperature Sensor

A PTC at the rectifier cooling rail is used to detect its temperature. The signal is routed to the D110 Power-UP board where a detection logic reports a cooling error in case the temperature exceeds 60 °C.

4.3.1.3 AC Voltage Monitoring

An AC voltage monitoring circuit checks the secondary output voltages of the main transformer T1 to be within specifications (283 V_{eff.} +/- 10%). If ok, the green LED V185 at the board lights up. The monitoring signals Volt_ok1(2, 3) are fed to the D110 additionally (see D110 description).

4.3.1.4 DC Voltage Monitoring

The D100 rectifier board is equipped with a built-in over (>450 VDC) and under voltage (<80 VDC) monitoring. These errors are monitored and fed via D110 and D70 to the D40 Modulator board where they are monitored. To indicate the presence of the DC voltage, five yellow LEDs (V1017-V1021) light up as long as the voltage level is higher than 80 V.

4.3.2 LEDs

Tab. 16 D100 LEDs

LED	Status	Description
V185 (green)	On	all secondary AC voltages are ok (283 V _{eff.} +/- 10 %)
V1017- V1021 (yellow)	On	DC voltage present(>80 V)

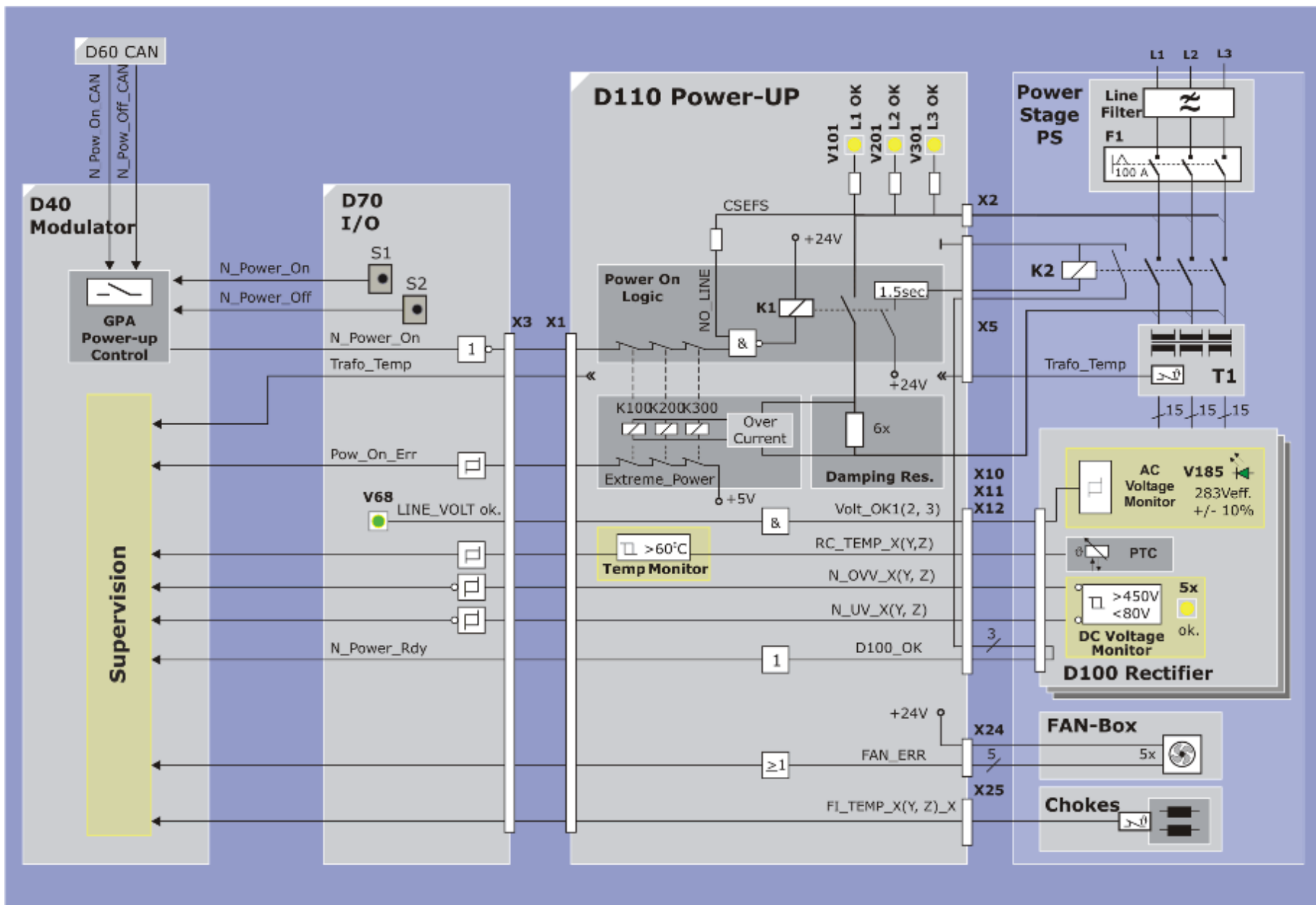
4.4 D110 Power-UP

4.4.1 Overview

The D110 is responsible for the following main functions:

- **Power switching of main transformer T1** (soft start)
- **Overload detection** for current limiting resistors
- **Temperature monitoring**

Fig. 26: D110 Power-UP



4.4.2 Function

4.4.2.1 GPA Power-on

The power switching is initiated by the N_POW_ON_CAN signal from the D60 or with push button S1 on the D70.

The D40 first performs a test to check for a "Power Stage cable defect" (see D40 description). If successful, the N_POWER_ON signal is set and a "Transformer soft start" is performed.

4.4.2.2 Transformer Soft Start

The Power-on logic on the D110 energizes contactor K1, feeding the mains voltage to the transformer via current limiting resistors. After a delay of 1.5 seconds (and no errors are detected) contactor K2 is enabled to establish a direct connection to the mains supply.

A help contactor on K2 is used to inform the modulator that the power-up was successful (N_POWER_RDY). This signal is also looped over X10, X11, X12 to the three D100 Rectifier boards to assure they are plugged in (D100_OK signal). All modulators and regulators are now enabled and the D40 activates its DC low-voltage monitoring (N_UV_X(Y, Z) signals).

4.4.2.3 Overload Detection

If the line voltage is missing or a fault occurs, relays K1 and K2 will not be activated, or are deactivated if already on. During power-up the current via the current limiting resistors is monitored. In case of overload, the respective relay (K100, K200, K300) is activated and interrupts the K1 control voltage. The "EXTREME_POWER" error is generated and distributed to the D40 Modulator board (via D70).

4.4.2.4 GPA Power-off

In case of an error the GPA can be switched off by the D60 (N_POW_OFF_CAN signal). Push button S2 on the D70 board can also be used (N_POW_OFF_SW). Both signals are fed to the D40 where a "soft stop" blocks all modulator axis, the Power Stage low-voltage monitoring is disabled and the N_POWER_ON signal is set to high logic status (relay K1 and K2 opens). The modulators are finally switched off by "hard stop" (see D40 description).

4.4.2.5 Fan Monitoring

A feedback voltage (low = ok) from the five fans of the Power Stage and choke-section is used to check the proper function. If one of the signals fails, a FAN_ERR is reported via D70 to the D40 Modulator board.

4.4.2.6 Rectifier Voltage Monitoring

Each of the three D100 boards is equipped with monitoring circuitry for the secondary AC-voltages from the main transformer T1. If all voltages are in spec. (283 VAC +/- 10%), the Voltage_OK1(2, 3) signal is generated by the D110 and fed to the D70 board (a green LED V68 lights up if all AC voltages are ok, but the D60 CAN & Service board does not report that status). The D100 rectifier boards are equipped with an over- (>450 VDC) and under voltage (<80 VDC) monitoring. The corresponding error signals N_OVV_X(Y, Z) and N_UV_X(Y, Z) are routed via the D70 to the D40 Modulator board (see D40 and D110 description).

4.4.2.7 Temperature Monitoring

The D110 monitors the following temperatures:

- **Main transformer temperature monitoring:** The main transformer T1 is protected with an over-temperature switch that opens at 180 °C (TRAFO_TEMP). The signal is sent via the D70 to the D40 Modulator board.
- **Rectifier temperature monitoring:** A PTC placed on each of the rectifier heat-sinks is monitored for temperature. An error is reported via D70 to the D40 board when exceeding 60 °C.
- **Choke temperature monitoring:** The output filter chokes of the Power Stages are equipped with temperature switches. When they open, the FI_TEMP_X(Y, Z) informs the D40.

4.4.3 LEDs

Tab. 17 D110 LEDs

LED	Status	Description
V101	on	line voltage present (phase L1)
V201	on	line voltage present (phase L2)
V301	on	line voltage present (phase L3)

5.1 Overview

Fig. 27: Power Stage Modules

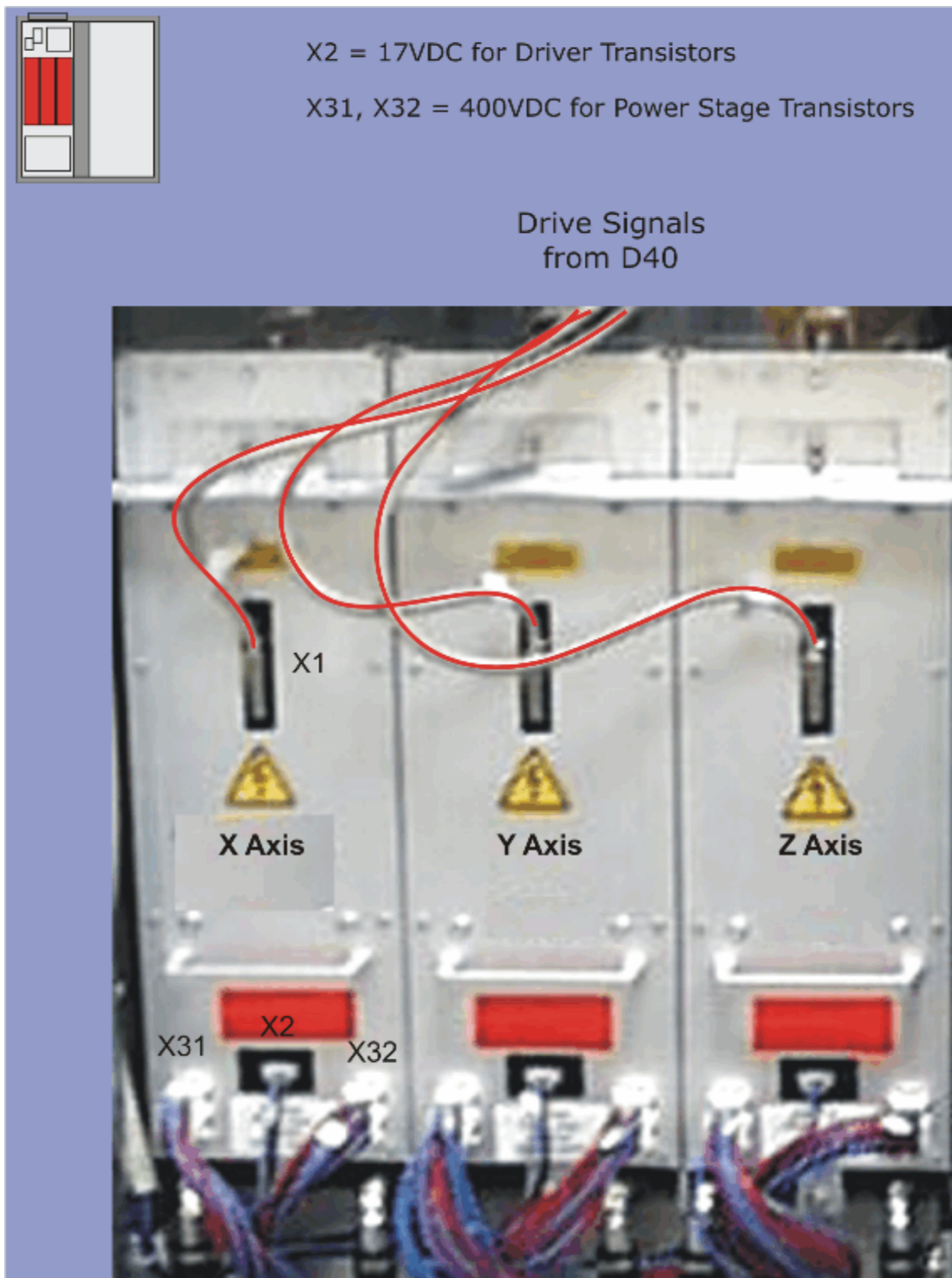
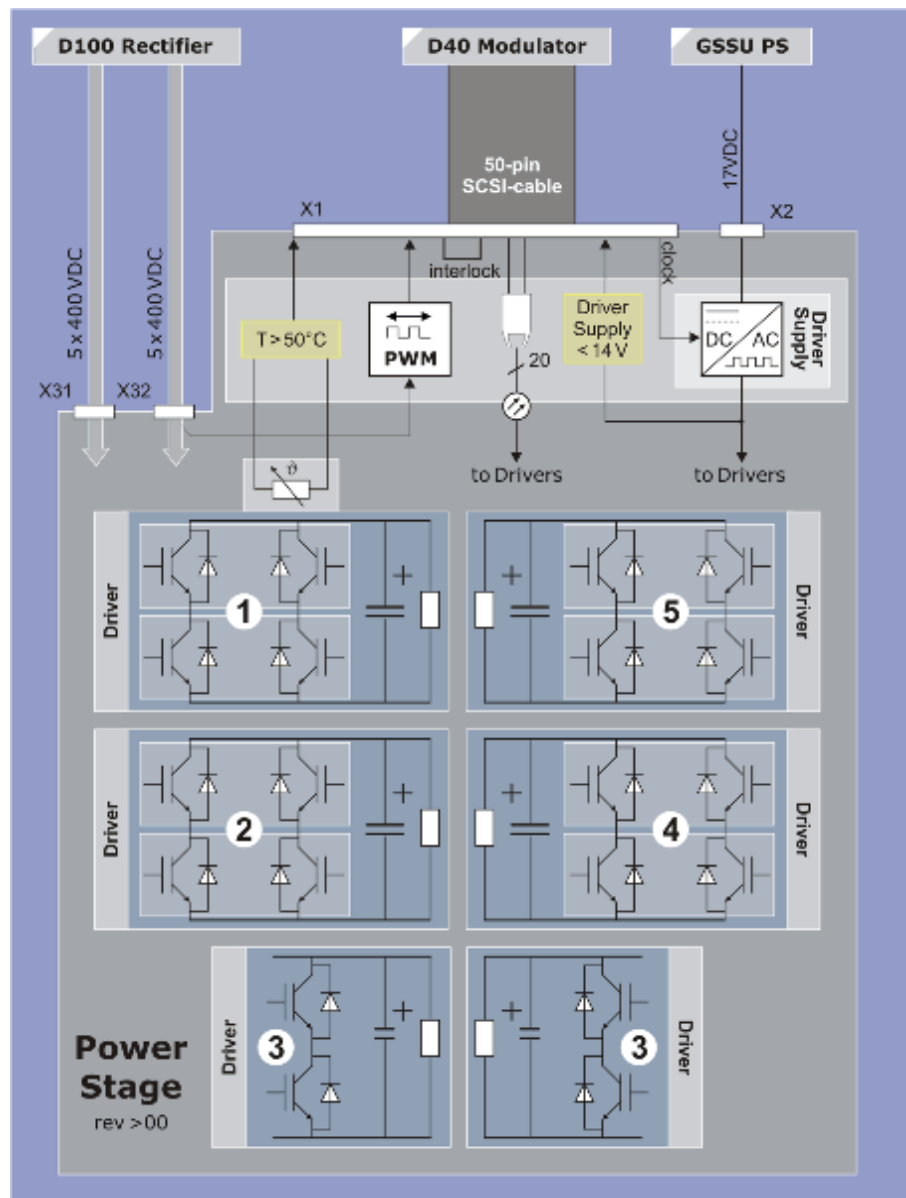


Fig. 28: One Power Stage Module



Each of the three Power Stage modules consists of 5 cascades (H-bridges) each with its own 400 VDC supply. The presence of this supply voltage is indicated by five yellow LEDs at each of the three D100 rectifier boards. Each Power Stage can deliver a maximum output of 650 A and 2000 V. The following components can be found inside:

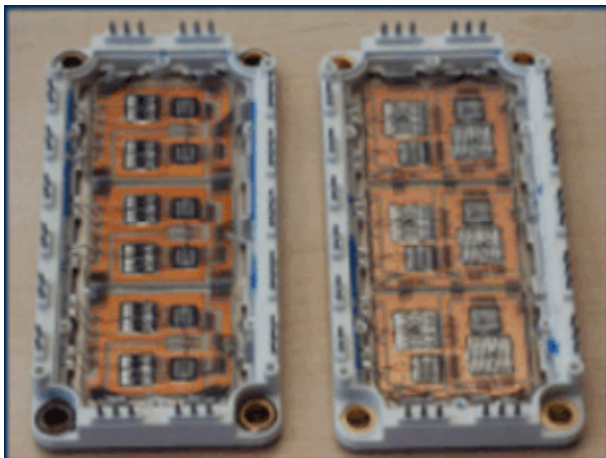
- **5 water cooled H-bridges** (cascades) with 4 IGBT switches each
- **5 times 400 VDC capacitor blocks**
- **Integrated driver supply** with galvanically isolated switch-mode power supply
- **Integrated opto-transmitter and opto-receiver** for drive and monitoring-signals



The Power Stage is a Field Replacable Unit (FRU)!

5.2 Function

Fig. 29: Two switches containing 6 IGBTs each

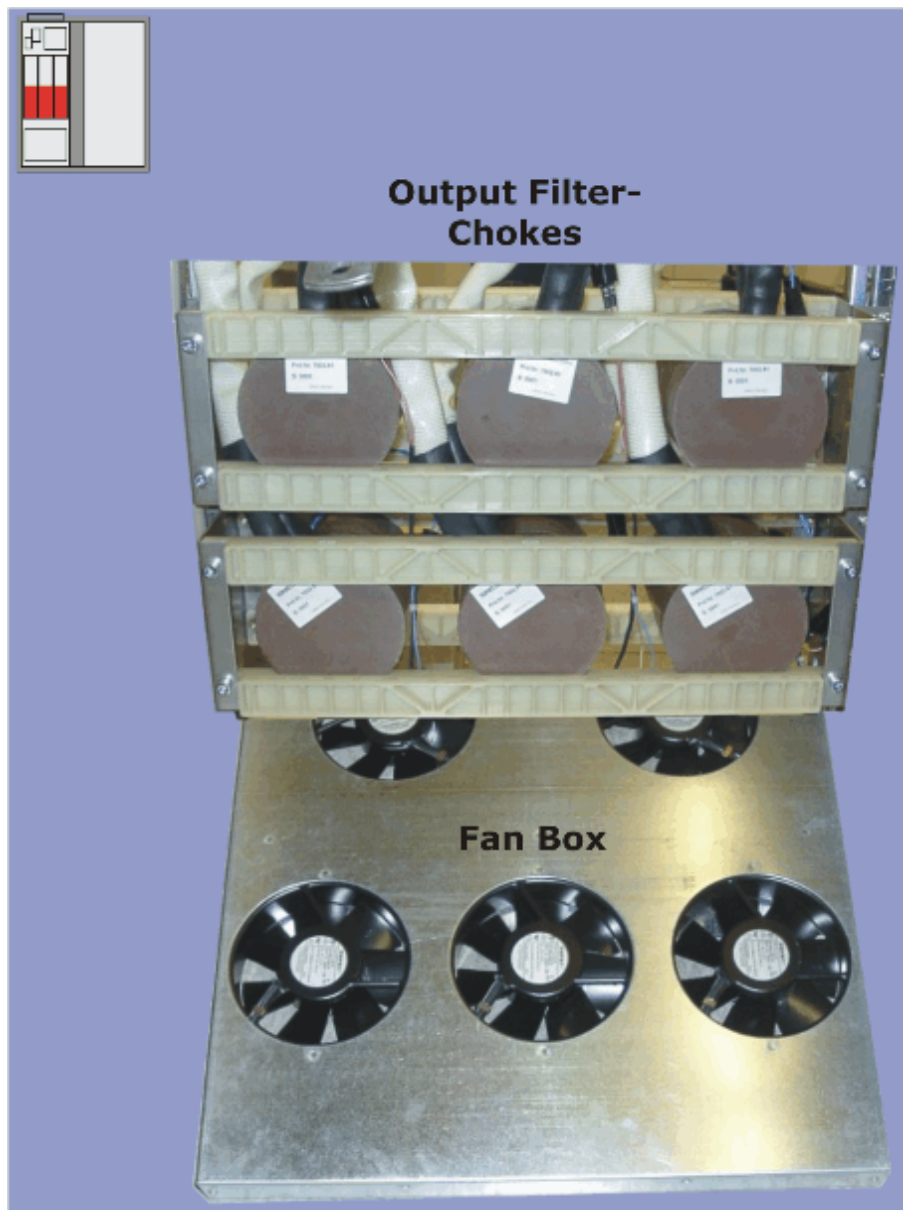


Each cascade consists of four switches. The switches are made from 6 parallel connected IGBTs (insulated gate bipolar transistor) with respective free-wheeling diodes anti-parallel.

The switching principle of each cascade is identical to the Power Stage switching described in the MR-Basics Gradient System documentation.

6.1 Overview

Fig. 30: Output Filter-Chokes and Fan Box



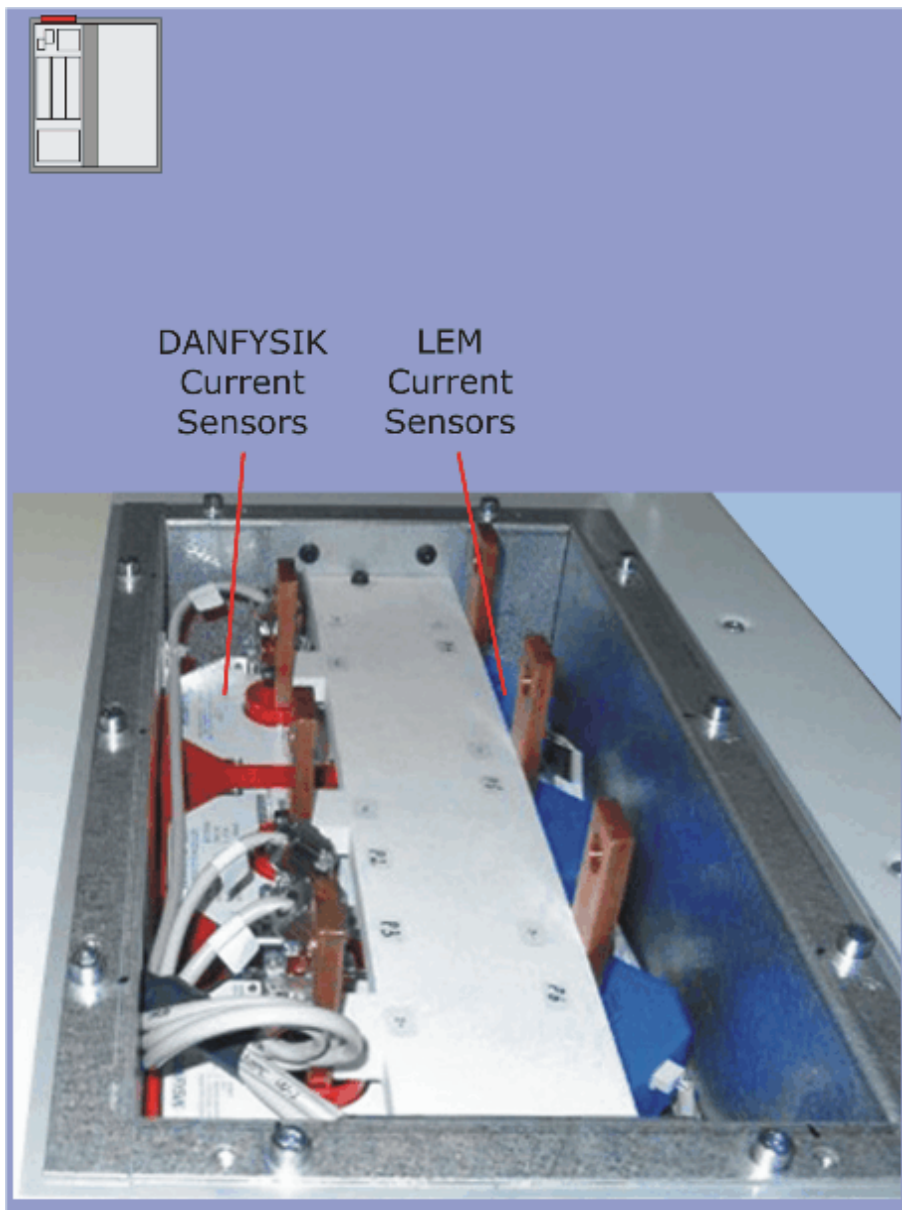
Six Output Filter-Chokes are located in a box (Choke-Set). The switching ripple on the gradient current is eliminated by the inductance of two filter-chokes per axis.

6.1.1 Over Temperature Protection

Each choke is protected by a thermal-switch that opens in case of over temperature. A corresponding FI_TEMPX(Y, Z) signal is generated and fed via D110 and D70 to the D40 Modulator-board.

7.1 Overview

Fig. 31: Current Converter Unit (CCU)



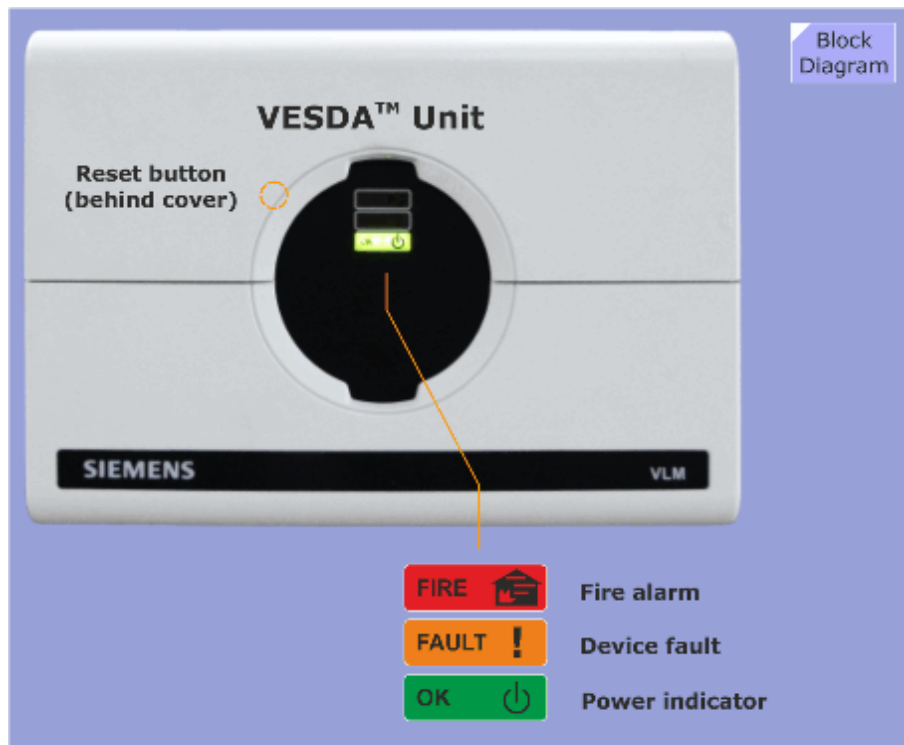
The Current Converter Unit consists of three DANFYSIK current sensors and three LEM current sensors. The sensors are placed in the supply and return sides of the gradient coil, measuring the current both going to and coming back from the coil. The **DANFYSIK** current sensors are used to supply precise actual current values needed for the regulation. The **LEM** current sensors are used only for monitoring purposes. In addition, filter capacitors are installed in this unit.



Each single current sensor is a Field Replacable Unit (FRU)!

8.1 Overview

Fig. 32: Gradient Supervision SIEMENS_VLM



MAGNETOM Trio a Tim Systems have an externally mounted **Gradient Supervision** unit called the **Siemens VLM** (a smoke detection product line) VESDA™ (product name). The unit incorporates a very sensitive laser-based light scatter detection system which can detect even small amounts of smoke particles emitted from overheated connections. The purpose of this unit is to provide continuous monitoring of all critical gradient connections.

Poorly fastened gradient cable connections have higher contact resistance which will lead to connector/cable overheating and damage and, in worst case, even fire.

The unit is integrated into the MR system. In case of a significant smoke particle concentration, a first relay is opened which will be processed by the AMC and an error message (evacuate patient) is posted to the monitor. One minute later, a second VLM relay will open and the power distributor will be forced to switch-off the MR system to stand by mode.

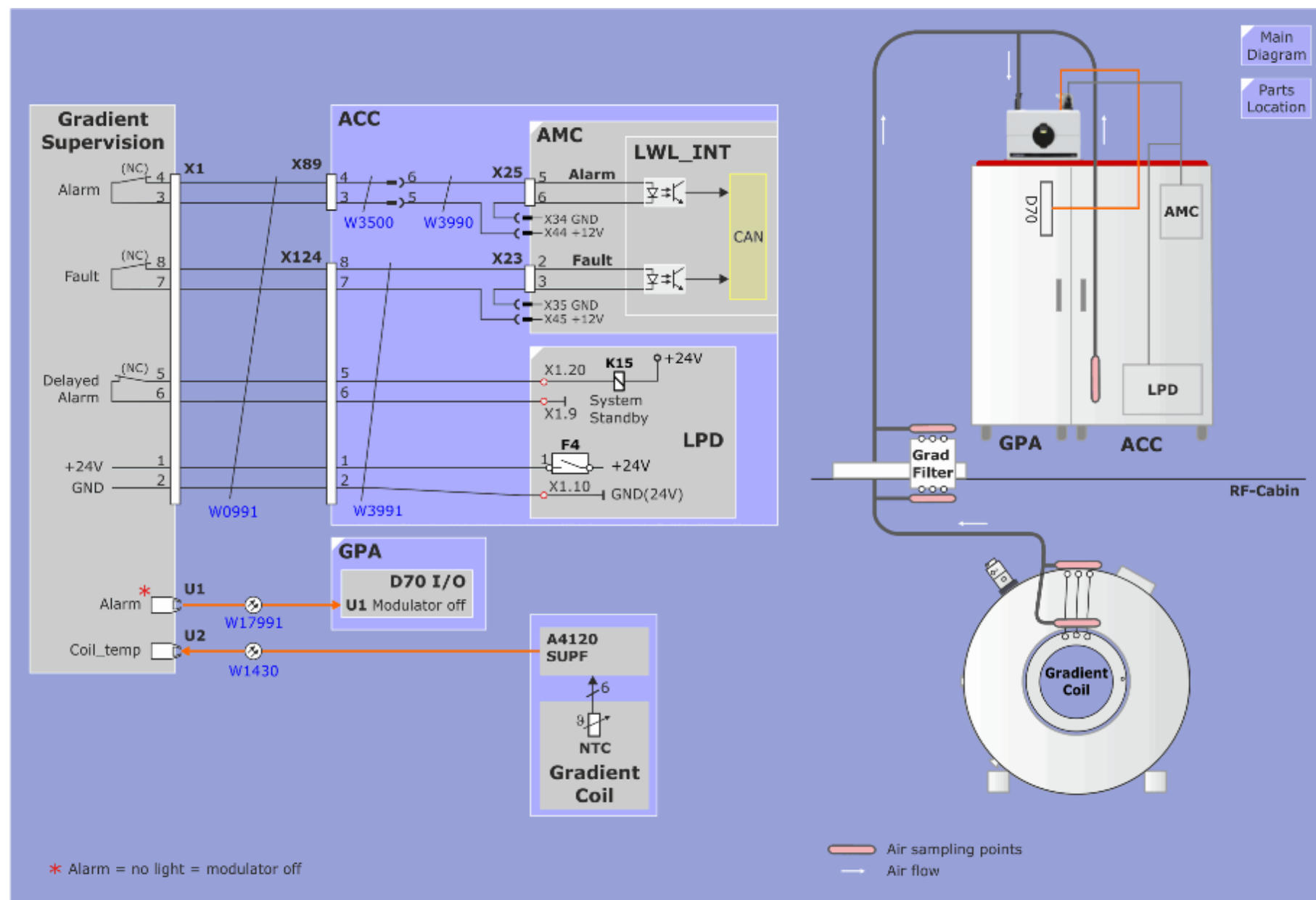
The Gradient Supervision also provides a possibility to access stored data via a RS232 serial connection. With help of the optional software Medical VSC (contact HSC), the user can retrieve important data like progression of smoke particle concentration, alarm and warning history which may be helpful to analyze and understand fault scenarios.

8.2 Function

Air is continuously drawn through a hose system which is mounted over all critical gradient connections and in the ACC cabinet. The air samples are fed through a filter. The filtering of dust and dirt from the air sample is important in order to keep the units optical detector surfaces free from contamination to ensure accurate readings and a long detector life.

Once filtered, the air sample is evaluated by a laser detector. If the sample contains smoke particles (e.g. emitted by overheated gradient connections) the laser beam is scattered which is detected by a highly sensitive optical receiver system.

Fig. 33: Gradient Supervision Block Diagram



8.2.1 Power Supply

The unit is powered with +24 VDC supplied by the 24/36 VDC power supply in the LPD (connector X124 at EPC roof). The supply voltage can be switched with circuit breaker F4.

8.2.2 Device Calibration

8.2.2.1 Warm-up

After an initial power-up, the yellow FAULT LED lights-up for 5 seconds and the unit runs through a 90 second warm-up cycle.

8.2.2.2 Reference Airflow Calibration

After initial installation of the unit it will be necessary to **calibrate** the nominal value of the reference air.



For correct calibration, the sample hose system must be fully attached. Also, close RF-room door to set standard air pressure condition in the RF-cabin.

To start the calibration:

- The unit must be powered-up for more than 90 seconds (= completed warm-up)
- Press the units' RESET button under the cover for 10 seconds until all 3 LEDs start flashing
- Release RESET button, all LEDs go off
- LEDs flash again, press RESET once more to confirm setting
 - » The current air flow through the device (which depends on the individual system configuration and hose length) is now **set to 100%**.
 - » Deviations larger than +/-30 % (+/-20 % at older VLMs) from the 100% nominal air flow (e.g. a blocked or disconnected sample hose) will result in a **device fault**.

8.2.3 Device Fault

8.2.3.1 Fault LED

In case of a malfunction the units **yellow FAULT LED** will light.

In fault mode, the supervisions' sensitivity is potentially affected, but the operation of the MR scanner is still possible. A constant light indicates a major fault, a flashing LED a minor problem.

The following list contains some possible causes for a device fault:

- **Wrong air flow** - more than +/-30 % (+/-20 % at older VLMs) deviation to the calibrated 100% reference air-flow value.
- **Excessive differential pressure** - more than 50 Pascal (0.5 mbar) between RF-cabin in-/ outside. This problem can be caused e.g. by an air condition system and will result

in the above mentioned air flow deviation of more than +/- 30 % (+/-20 % at older VLMs). A possible solution is, to route an equalization tube from the units exhaust back into the RF-cabin, thus equalizing the pressure difference seen by the VLM.

- **Defective laser scatter light detector**
- **Clogged VLM air filter**



The differential pressure between RF-cabin in-/ outside should not exceed 50 Pascal (0.5 mbar).

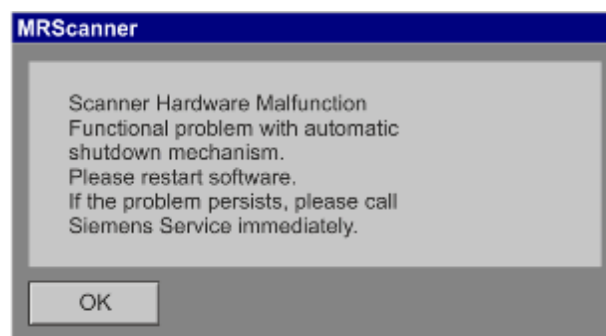
Otherwise, the VLM may report air flow device faults once the RF-room door is opened.

Ensure, the differential pressure is less than 50 Pascal (0.5 mbar) or install a pressure equalization tube from the units' exhaust back into the RF-cabin.

8.2.3.2 Device Fault pop-up

The customer is informed with a pop-up window.

Fig. 34: Device Fault pop-up



8.2.3.3 Device Fault Reset

A Device Fault will be **automatically reset** when the fault condition is not present anymore.

8.2.4 Fire Alarm

8.2.4.1 FIRE LED

In case smoke gases are detected within the monitored areas, the SIEMENS_VLM sets the **red FIRE LED**.

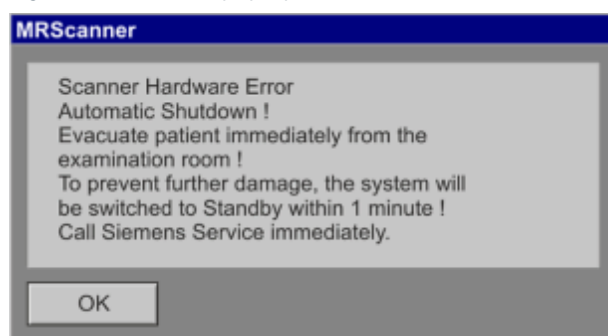
Fig. 35: Fire Alarm LED



8.2.4.2 Alarm pop-up

The customer is informed about a fire alarm with a pop-up window.

Fig. 36: Fire Alarm pop-up



8.2.4.3 Measurement Abort

The MR system is informed about the critical condition via the connection to the LWL_INT (cable W0991). Consequently, a running measurement is aborted and the modulators of the GPA are blocked (via FOC W17991).

8.2.4.4 Power Switch-off

After **one minute** the MR system is **switched to stand-by**. The signal is fed via X1/ pin 5, 6 VLM to X124/ pin 5, 6 to ACC roof.

8.2.4.5 Fire Alarm Reset

A Fire Alarm condition is latched by the SIEMENS_VLM to inhibit further system power switch-on attempts.

Once, the smoke concentration went below its alarm threshold, a CSE can reset the fault by pressing the units' **RESET button**. It can be found under the units' cover left to the indicator LEDs.

8.3 Jumpers

Jumper	Pins	Setting	Location	Description
Config	2-3	closed	under the units' cover, left of the FOC connector	no alarm = FOC light on alarm = FOC light off

8.4 Switches

Switch	Location	Description
RESET	under the units' cover, left of the control LEDs	■ Resets latched faults ■ Set-up button for device calibration

8.5 LEDs

LED	ON	FLASHING	Description
OK (green)	Power on	n.a.	Detector is powered up and in normal operation
FAULT (yellow)	Major Fault	Minor Fault	Call service, most common faults: ■ Air flow deviation more than 30 % (20 % at older VLMs) from calibrated 100 % nominal flow, see (→ Device Fault / Page 68). ■ Calibration of airflow required. ■ Defective laser scatter light detector. ■ Clogged VLM air filter. Note: Several additional, less probable faults exist. VLM faults can be further diagnosed with a service PC at the RS 232 serial interface and optional software "Medical VSC" (contact HSC).
FIRE (red)	Smoke detected	n.a.	■ Alarm pop-up generated ■ GPA modulator will be disabled ■ MR-scanner will be switched off to stand-by 1 minute later

8.6 Service Procedures

The function of the Gradient Supervision must be checked every 12 months during maintenance.



For details please refer to the Safety-Related Tests Procedure in the CB-DOC.

Version 02:

- » New section (→ Gradient Supervision / Page 65) added.

There are no Hazard IDs in this document.

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